Fundamentals of Artificial Intelligence

Weekly assignment screenshots

40 assignment sets | 113 source images | Collected 21 October 2026

Week	2026	2025	2024	2023	2022	2019
01	p. 2	p. 5	-	p. 7	p. 11	p. 16
02	p. 17	p. 22	-	-	p. 25	p. 30
03	p. 31	p. 35	-	-	-	p. 38
04	p. 39	p. 42	-	-	-	p. 47
05	p. 48	p. 51	-	-	-	p. 54
06	p. 55	p. 58	p. 63	-	-	p. 66
07	p. 67	p. 71	-	-	-	p. 74
08	p. 75	p. 77	-	-	-	p. 81
09	p. 82	p. 86	-	-	-	p. 90
10	p. 91	p. 94	-	-	-	p. 98
11	p. 99	p. 102	-	-	-	p. 106
12	p. 107	p. 110	-	-	-	p. 114

Click a page number. A dash means no screenshots were collected for that year/week.

2026 and 2019: official NPTEL sources. Other years: publicly available copies.

Original answer keys, highlights, and selections are preserved; public answers are unverified.

Known source error: 2022 Week 1, Q9 is A (Weak AI); that webpage prints B.

These are weekly assignments, not authenticated final-exam papers.

Source links appear on every image page. Individual images are in screenshots/Week_01 through Week_12.

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Fundamentals of Artificial Intelligence
MOOCs; July – Dec 2026
Indian Institute of Technology Guwahati
Assignment No. 1: Detailed Solution

10 Marks

Each question carries 01 Mark each.

Q1. Who coined the term Artificial Intelligence during the Dartmouth Conference in 1956?

A. Marvin Minsky
B. Alan Turing
C. John McCarthy
D. Claude Shannon

Correct Answer: (C)

Q2. The Turing Test or Imitation Game was proposed as a simple test that could be used to _____.

A. determine whether a machine can exhibit intelligent behavior indistinguishable from that of a human.
B. prove that machines are conscious.
C. measure the computational speed of a machine.
D. demonstrate that machines can replace human intelligence.

Correct Answer: (A)

Q3. A production system consists of _____

A. Facts, Rules, and Control Strategy
B. Data, Compiler, and Operating System
C. Database, SQL, and Queries.
D. Constraints

Correct Answer: (A)

Q4. Which of the following best characterizes an AI Technique?

A. Uses exhaustive computation only.
B. Exploits knowledge that captures generalizations and can be easily modified.
C. Solves only numerical optimization problems.
D. Depends entirely on faster processors.

Correct Answer: (B)

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Q5 The 8-puzzle problem has a total of _____ possible configurations (states) in its state space.

A. 9!
B. 8!
C. Infinite
D. 64!
Correct Answer: (A)

Q6. In a specialized production system, the order in which applicable rules are applied to the database is unimportant in a _____ production system.

A. Decomposable
B. Commutative
C. Specialized
D. AND/ OR
Correct Answer: (B)

Q7. Uninformed search is also known as _____.

A. Blind Search
B. Heuristic Search
C. Knowledge-Based Search
D. Best-First Search
Correct Answer: (A)

Q8. In state-space search, the state space refers to

A. the shortest path between the start state and the goal state.
B. the set of all possible configurations of the relevant objects.
C. only the goal states reachable from the initial state.
D. the collection of search algorithms used to solve the problem.
Correct Answer: (B)

Q9. In Breadth-First Search (BFS), which of the following statements is correct?

A. The fringe is maintained as a FIFO queue.

B. The fringe is maintained as a LIFO stack.

C. The deepest unexpanded node is expanded first.

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E.Newly generated successor nodes are added to the front of the OPEN list.

Correct Answer: (A)

Q10. Which of the following statements is FALSE regarding Bidirectional Search?

A. It requires the ability to generate predecessor states.

B. It is always an optimal search strategy.

C. The search terminates when the two search frontiers intersect.

D. It works best when there is a unique start state and a unique goal state.

Correct Answer: (B)

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Fundamentals of Artificial Intelligence
MOOCs; July - Dec 2025

Assignment No. 1

10 Marks

Each question carries 01 Mark each.

Q1. George B. Dyson refers to _____ as the patriarch of artificial intelligence in his book, "Darwin Among the Machines: The Evolution of Global Intelligence".

A. John McCarthy
B. Thomas Hobbes
C. Alan Turing
D. Herbert Simon

Q2. Heuristic searching, which is now a central idea of AI was employed to design _____.

A. The Logic Theorist, considered by many to be the first AI program.
B. machines with minds, in the full and literal sense.
C. electromechanical machines to read thousands of German radio intercepts.
D. a simple test that could be used to prove that machines could think.

Q3. Of the different dimensions of Artificial Intelligence, "Think Rationally" is to
A. model human cognition.
B. formalize the inference process.
C. do the right thing.
D. exhibit human behaviour.

Q4. _____ aims at building machines that act intelligently, without taking a position on whether or not the machines actually are intelligent.
A. Weak AI
B. Strong AI
C. Connectionist AI
D. Symbolic AI

Q5 Assertion A AI Production System with uninformed control system have Low Control Strategy Cost.
Reason R At the uninformed extreme, the control strategy is guided by the problem knowledge great enough to select a correct rule every time.

Mark the correct choice as
A. Both A and R are true and R is the correct explanation for A
B. Both A and R are true but R is not the correct explanation for A
C. A is True but R is False
D. A is false but R is True

Q6. Fringe are the nodes waiting in a sequence to be explored. Fringe for the Depth-First Search is a _____ data structure.
A. Unordered.
B. FIFO.
C. LIFO.

Public-copy highlights reproduced as supplied; answers unverified.

D. Queue.

Q7. Fringe are the nodes waiting in a sequence to be explored. Fringe for the Breadth-First Search is a _____ data structure.

A. Unordered.
B. FIFO.
C. LIFO.
D. Stack.

Q8. There are two distinct specialized production systems; Order in which a set of applicable rules is applied to a database is unimportant for a _____ production system.

A. decomposable
B. commutative
C. specialized
D. AND/ OR

Q9. For bidirectional search is a graph search algorithm that runs two simultaneous searches. Which of the following statements is false for bidirectional search?

A. Requires the ability to generate *predecessor* states.
B. Bidirectional search is *always* optimal.
C. Stop when the frontiers intersect.
D. Works well only when there are unique start and goal states.

Q10. Identify the criterion on which search strategies are evaluated:

A. completeness
B. consistency
C. space complexity
D. admissibility

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Assignment 1

The due date for submitting this assignment has passed.

Due on 2023-08-09, 23:59 IST.

As per our records you have not submitted this assignment.

1) George B. Dyson refers to _____ as the patriarch of artificial intelligence in his book, "Darwin Among the Machines: The Evolution of Global Intelligence". ***1 point***

- A. John McCarthy
- B. Thomas Hobbes
- C. Alan Turing
- D. Herbert Simon

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Thomas Hobbes

2) Heuristic searching, which is now a central idea of AI was employed to design _____. ***1 point***

- A. The Logic Theorist, considered by many to be the first AI program.
- B. machines with minds, in the full and literal sense.
- C. electromechanical machines to read thousands of German radio intercepts.
- D. a simple test that could be used to prove that machines could think.

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. electromechanical machines to read thousands of German radio intercepts.

3) Production systems can be defined as a kind of cognitive architecture, in which knowledge is represented in the form of rules. Which of the following is not an essential component of an AI production system? ***1 point***

- A. Global Database
- B. Set of Rules
- C. Learning element
- D. Control Strategy

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. Learning element

4) Transforming a problem statement into the components of a production system is called the _____ in AI. ***1 point***

- A. mapping problem
- B. translation problem
- C. representation problem
- D. generation problem

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. representation problem

5) Assertion A AI Production System with uninformed control system have Low Control Strategy Cost. ***1 point***

Reason R At the uninformed extreme, the control strategy is guided by the problem knowledge great enough to select a correct rule every time. Mark the correct choice as

- A. Both A and R are true and R is the correct explanation for A
- B. Both A and R are true but R is not the correct explanation for A
- C. A is True but R is False
- D. A is false but R is True

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. A is True but R is False

6) Assertion AAn irrevocable control regime can always be used in a commutative production system. Reason R Applying an inappropriate rule in a commutative production system delays, but never prevents termination. Mark the correct choice as ***1 point***

- A. Both A and R are true and R is the correct explanation for A
- B. Both A and R are true but R is not the correct explanation for A.
- C. A is True but R is False
- D. A is false but R is True

No, the answer is incorrect.

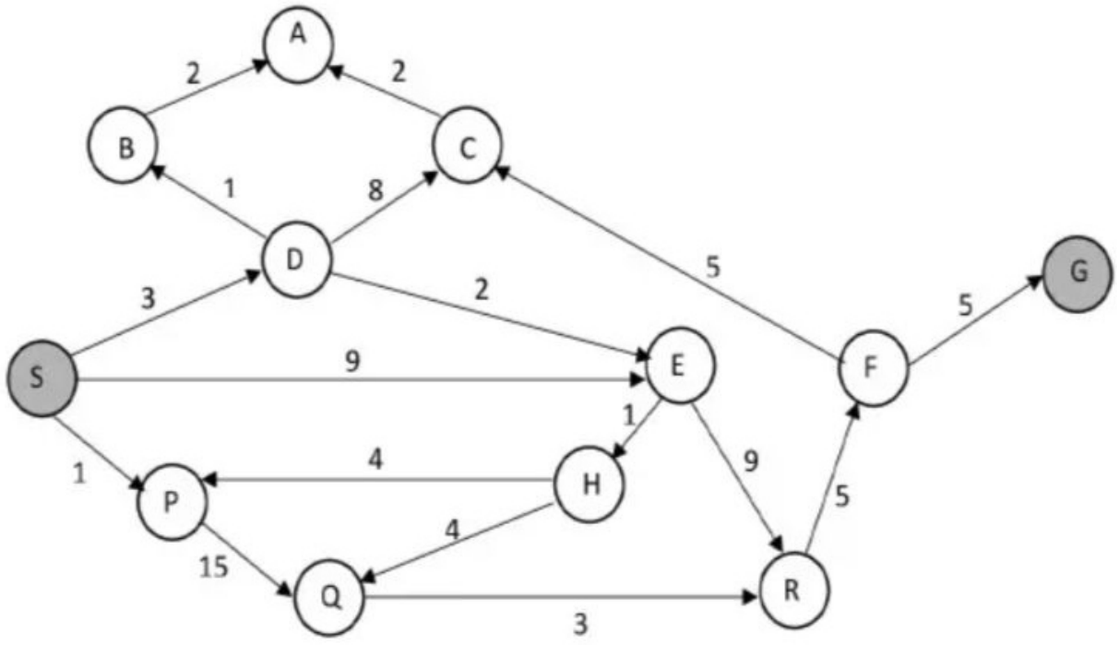
-

Score: 0

Accepted Answers:

B. Both A and R are true but R is not the correct explanation for A.

For Q7 – Q10: Consider the graph shown below. Start is marked S and goal is marked G. Cost of each edge is marked.



7) Mark the nodes expanded in second level (i.e., 2-steps from the start node) for Breadth-First Search. ***1 point***

- A. Q; R; E; C; B.
- B. Q; R; H; C; B.
- C. Q; H; E; C; B.
- D. Q; R; H; E; B.

No, the answer is incorrect.
Score: 0

Accepted Answers:

B. Q; R; H; C; B.

8) For Breadth-First Search of the above graph, the solution path found is _____. ***1 point***

- A. S-E-H-Q-R-F-G
- B. S-D-E-R-F-G
- C. S-E-R-F-G
- D. S-P-Q-R-F-G

No, the answer is incorrect.
Score: 0

Accepted Answers:

C. S-E-R-F-G

9) For Breadth First Search, the following statements are correct: ***1 point***

- A. Fringe is a FIFO queue.

- B. Fringe is a LIFO queue.
- C. Expand shallowest unexpanded node first.
- D. Nodes are added to the front of the OPEN list.

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Fringe is a FIFO queue.
C. Expand shallowest unexpanded node first.

10) When a priority queue is used instead of a simple queue in a Breadth First Search, we insert nodes in the increasing order of the cost of the path so far. This is the Uniform-cost Search algorithm. Find the cost of the solution path found using Uniform-cost Search. ***1 point***

- A. 24
- B. 28
- C. 23
- D. 29

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. 23

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NPTEL Fundamentals of Artificial Intelligence Assignment 1 Answers 2022 [July-Dec]

Q1. Aristotle was one of the firsts to attempt to codify "thinking". His _____ provided patterns of argument structure that always gave correct conclusions, given correct premises

A. deductions
B. formalism
C. syllogisms
D. axioms

Answer:- c

2. Turing test or Imitation Game, was put forth as a simple test that could be used to _____

A. prove machines could think.
B. break the enigma code.
C. show human are intelligent.
D. show machines can compute.

Answer:- a

3. Overall computational cost of a AI Production System is in two major categories: Rule Application Cos and Control Strategy Cost. For an Uninformed Control System, we have the following:

A. Low Rule Application Cost and High Control Strategy Cost
B. Low Rule Application Cost and Low Control Strategy Cost
C. High Rule Application Cost and Low Control Strategy Cost
D. High Rule Application Cost and High Control Strategy Cost

Answer:- c

4. For more complex games, such as chess or checker the AND/OR search to termination is out of question. Which of the following statements are true?

I. Our goal in searching such a game tree might be, instead, merely to find a good first move.
II. Extract from the search graph an estimate of the 'best' first move.

A. Both I and II.
B. Only II
C. Only I
D. Both I and II are false.

Answer:- a

5. For Breadth First Search, the following statements are correct

A. Fringe is a FIFO queue
B. Fringe is a LIFO queue
C. Deepest node is expanded first.
D. Nodes are added to the front of the OPEN list.

Answer:- a

6. Bidirectional search is a graph search algorithm that runs two simultaneous searches. Which of the following statements is false for bidirectional search? .

A. Requires the ability to generate predecessor states.
B. Bidirectional search is always optimal.
C. Stop when the frontiers intersect.
D. Works well only when there are unique start and goal states

Answer:- b

7. There are two distinct specialized production systems; Order in which a set of applicable rules is applied to a database is unimportant for a _____ production system.

A. Decomposable
B. Commutative
C. Specialized
D. AND/ OR

Answer:- b

8. To solve a problem using a production system, one must specify _____

A. The Global database, Rules and Control Strategy
B. Some search technique
C. Initial facts only
D. Constraints

Answer:- a

9. _____ aims at building machines that act intelligently, without taking a position on whether or not the machines actually are intelligent.

A. Weak AI
B. Strong AI
C. Connectionist AI
D. Symbolic AI

Answer:- b

10. The AO* algorithm can best be seen as the following TWO major operations:

A. Step I: Top-down: graph-growing; Step II: Bottom-up: cost-revising, SOLVE-labelling.
B. Step I: Bottom-up: graph-growing; Step II: Top-down: cost-revising, SOLVE-labelling.
C. Step I: Top-down: graph-growing, SOLVE-labelling; Step II: Bottom-up: cost-revising.
D. Step I: Bottom-up: graph-growing, SOLVE-labelling; Step II: Top-down: cost-revising.

Answer:- a

Q9 correction: A (Weak AI). The source webpage prints B.

Assignment 1

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2019-08-14, 23:59 IST.

1) Who coined the termed Artificial Intelligence? ***1 point***

- A. Marvin Minsky
- B. Alan Turing
- C. John McCarthy
- D. Issac Asimov

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. John McCarthy

2) Turing test or Imitation Game, was put forth as a simple test that could be used to _____ ***1 point***

- A. Prove machines could think.
- B. Break the enigma code.
- C. Show human are intelligent.
- D. Show machines can compute.

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Prove machines could think.

3) To solve a problem using a production system, one must specify _____ ***1 point***

- A. The Global database, Rules and Control Strategy
- B. Some search technique
- C. Initial facts only
- D. Constraints

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. The Global database, Rules and Control Strategy

4) Overall computational cost of a AI Production System is in two major categories: Rule Application Cost and Control Strategy Cost. For an Uninformed Control System, we have the following: ***1 point***

- A. Low Rule Application Cost and High Control Strategy Cost
- B. Low Rule Application Cost and Low Control Strategy Cost
- C. High Rule Application Cost and Low Control Strategy Cost
- D. High Rule Application Cost and High Control Strategy Cost

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. High Rule Application Cost and Low Control Strategy Cost

5) The 8-puzzle problem have a total of _____ states in its State Space. ***1 point***

- A. 9!
- B. 8!
- C. Infinite
- D. 64!

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. 9!

6) There are two distinct specialized production systems; Order in which a set of applicable rules is applied to a database is unimportant for a _____ production system. ***1 point***

- A. Decomposable
- B. Commutative
- C. Specialized
- D. AND/ OR

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Commutative

7) Domain knowledge is used to make preferential choice between the child nodes in _____ ***1 point***

- A. Depth First Search
- B. Breadth First Search
- C. Heuristic Search
- D. Depth Limited Search

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. Heuristic Search

8) _____ aims at building machines that act intelligently, without taking a position on whether or not the machines actually are intelligent. ***1 point***

- A. Weak AI
- B. Strong AI
- C. Connectionist AI
- D. Symbolic AI

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Weak AI

9) For Breadth First Search, the following statements are correct ***1 point***

- A. Fringe is a FIFO queue
- B. Fringe is a LIFO queue
- C. Deepest node is expanded first.
- D. Nodes are added to the front of the OPEN list.

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Fringe is a FIFO queue

10) For bidirectional search is a graph search algorithm that runs two simultaneous searches. Which of the following statements is false for bidirectional search? ***1 point***

- A. Requires the ability to generate predecessor states.
- B. Bidirectional search is always optimal.
- C. Stop when the frontiers intersect.
- D. Works well only when there are unique start and goal states.

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Bidirectional search is always optimal.

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Assignment No. 2: Detailed Solution

10 Marks

Each question carries 01 Mark each.

Q1. The principal gain from using a heuristic function in AI search is the _____.

A. elimination of the goal test.
B. reduction of the search space.
C. generation of all successor states.
D. guarantee of finding the optimal solution.
Correct Answer: (B)

Q2. Heuristics obtained as solutions to relaxed problems are generally _____ because adding new actions should not increase the solution cost.

A. consistent
B. pessimistic
C. admissible
D. inadmissible
Correct Answer: (C)

Q3. Assertion–Reason

Assertion (A): An admissible heuristic never overestimates the actual cost of reaching the nearest goal.
Reason (R): An admissible heuristic is optimistic because its estimated cost is less than or equal to the actual cost.

Choose the correct answer:
A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, but R is not the correct explanation of A.
C. A is true, but R is false.
D. A is false, but R is true.
Correct Answer: (A)

Q4. Which of the following statements regarding A* Search is correct?

A. A* uses only the estimated cost from the current node to the goal.
B. A* uses only the path cost from the start node to the current node.

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C. A* evaluates a node using the estimated total cost, f(n) = g(n) + h(n).

D. A* expands nodes without using an evaluation function.

Correct Answer: (C)

Q5. Assertion–Reason

Assertion (A): A* Search generates an optimal solution when h(n) is admissible for tree search and consistent for graph search.

Reason (R): An admissible heuristic never overestimates the actual cost to the goal, while consistency ensures that the estimated total path cost does not decrease along a path.

Choose the correct answer:

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, but R is not the correct explanation of A.

C. A is true, but R is false.

D. A is false, but R is true.

Correct Answer: (A)

Q6. Assertion–Reason

Assertion (A): Simulated Annealing may accept a successor state that is worse than the current state.

Reason (R): Accepting an inferior state with a certain probability may help the search escape from a local optimum.

Choose the correct answer:

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, but R is not the correct explanation of A.

C. A is true, but R is false.

D. A is false, but R is true.

Correct Answer: (A)

Q7 - Q10

Consider an instance of the 8-puzzle problem. We are given the state on the left side and we want to transform it into the state on the right.

Start

1	3	4
8		5
7	2	6

Goal

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1	2	3
8		4
7	6	5

- A problem with fewer restrictions on the actions is called a relaxed problem.
- The cost of an optimal solution to a relaxed problem is an admissible heuristic for the original problem.
- Two admissible heuristics are: (a) Hamming Distance h_1 and (b) Manhattan Distance h_2.

Q7. Compute the Hamming Distance of the start state for the given instance, h_1(start).

A. 5
B. 7
C. 6
D. 4

Correct Answer: (A)

Explanation: Tiles 2, 3, 4, 5, 6 are out of place. So, h_1(start) = 5.

Q8. Compute the Manhattan Distance of the start state for the given instance, h_2(start).

A. 5
B. 7
C. 6
D. 4
Correct Answer: (C)
Explanation: Contributions: 2 = 2; 3 = 1; 4 = 1; 5 = 1; 6 = 1; all others = 0. So, sum = 2 + 1 + 1 + 1 + 1 = 6.

Q9. Find the number of moves required to reach the goal from the start using the Manhattan Distance heuristic.

A. 5
B. 7
C. 6
D. 4
Correct Answer: (C)
Explanation:

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```
Move 0:
1 3 4
8 _ 5
7 2 6

Move 1:
1 3 4
8 2 5
7 _ 6

Move 2:
1 3 4
8 2 5
7 6 _

Move 3:
1 3 4
8 2 _
7 6 5

Move 4:
1 3 _
8 2 4
7 6 5

Move 5:
1 _ 3
8 2 4
7 6 5

Move 6:
1 2 3
8 _ 4
7 6 5
```

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Q10. Which of the two heuristics, Hamming Distance h_1 and Manhattan Distance h_2, is more efficient?

A. Hamming Distance h_1
B. Manhattan Distance h_2
C. Both are equally good.
D. Both are equally poor.

Correct Answer: (B)

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Assignment 2: Detailed Solution
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10 Marks

Each question carries 01 Mark each.

Q1. A* generates an optimal solution under which of the following conditions?
I. If h(n) is an admissible heuristic and the search space is a tree.
II. If h(n) is a consistent heuristic and the search space is a graph.

A. A* never generate an optimal solution.
B. I only
C. Both I and II
D. II only

Q2. Fringe are the nodes waiting in a sequence to be explored. Fringe for the Depth-First Search is a _____ data structure.

A. Unordered.
B. FIFO.
C. LIFO.
D. Queue.

Q3. Solving a constraint satisfaction problem on a finite domain is an/a _____ problem with respect to the domain size.

A. P complete
B. NP complete
C. NP hard
D. Domain dependent

Q4. A solution to a CSP is a complete assignment that satisfies all the constraints. Which of the two are true for a solution to a CSP?
I. An assignment that does not violate any constraints is called consistent or legal.
II. A complete assignment is one in which every variable is mentioned.

A. Neither I nor II
B. Both I and II
C. I only
D. II only

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Assignment 2: Detailed Solution
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Q5. Conflict-directed Backjumping is better than Simple Backjumping because:

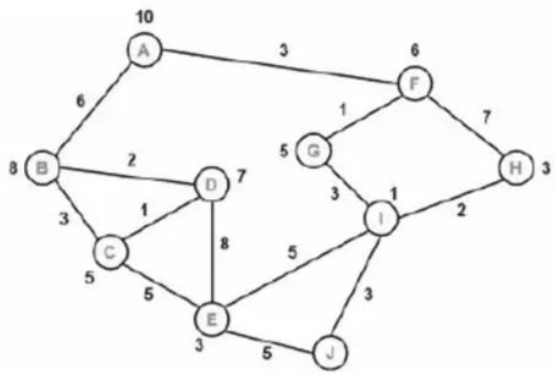
I. When a partial solution cannot be executed, the algorithm backtracks to the latest choice point.

II. When a contradiction occurs, remember the minimum set of variables from the conflict set that was responsible for the problem.

A. Neither I nor II
B. Both I and II
C. I only
D. II only

For Q6. – Q10.

Consider the following graph and A* Algorithm. The nodes are marked by upper case letters; the numbers written by the nodes represent their heuristic value. Node J is the goal node with heuristic value 0. The numbers on the edge is the cost of traversing that particular edge i.e., distance between the nodes.



Q6. The A* search algorithm is based on _____.

A. Breadth-First-Search
B. Depth-First –Search
C. Best-First-Search
D. Hill climbing

Q7. Starting with node A; node B and node F can be reached from node A. Which node does algorithm A* considers next? Why? [Here f(n) is the heuristic function at node n.]

A. Node F; f(F) > f(B).

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B. Node B; f(B)> f(F).
C. Node B; f(B)<f(G).
D. Node F; f(F)<f(B).

Q8. Which node among the successors of node I along the most-effective path (i.e., nodes that can be reached from node I) have the least value for the evaluation function f(n)?

A. Node E
B. Node H
C. Node J
D. Node G

Q9. Which node among the successors of node I along the most-effective path (i.e., nodes that can be reached from node I) is least promising?

A. Node E
B. Node H
C. Node J
D. Node G

Q10. What is the most cost-effective path to reach goal state J from start state A using the A* Algorithm?

A. A-F-H-I-J
B. A-B-D-E-J
C. A-B-C-E-J
D. A-F-G-I-J

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NPTEL Fundamentals of Artificial Intelligence Assignment 2 Answers 2022 [July-Dec]

Q1. The A* search algorithm is based on _____

A. Breadth-First-Search
B. Depth-First –Search
C. Best-First-Search
D. Hill climbing

Answer:- c

2. A* generates an optimal solution under which of the following conditions?
Condition 1: If h(n) is an admissible heuristic and the search space is a tree.
Condition 2: If h(n) is a consistent heuristic and the search space is a graph.

A. A* never generate an optimal solution.
B. Condition 1 only
C. Condition 2 only
D. Both Condition 1 and Condition 2

Answer:- d

3. Which of the following statements are true for inadmissible heuristics?

A. Overestimate the cost.
B. Breaks optimality by trapping good plans on the fringe.
C. Slow down search by assigning a lower cost to a bad plan.
D. Underestimate the cost.

Answer:- a, b, c

4. A solution to a CSP is a complete assignment that satisfies all the constraints. Which of the two are true for a solution to a CSP?

I. An assignment that does not violate any constraints is called consistent or legal.
II. A complete assignment is one in which every variable is mentioned.

A. Neither I nor II
B. Both I and II
C. I only
D. II only

Answer:- b

5. Arc Consistency is a method of constraint propagation. Which of the following statements are correct for arc consistency?

A. Arc consistency is substantially stronger than forward checking.
B. Detects failure early!
C. Arc consistency is one of the most powerful propagation techniques for binary constraints.
D. Provides a fast method of constraint propagation.

Answer:- b, c, b

6. _____ is a backtracking algorithm for CSP that is aware of the underlying constraint graph; determines where to jump back to, based on the actual conflict that it has recorded.

A. Chronological Backtracking
B. Conflict-directed Backjumping
C. Arc Consistency
D. Plain Backtracking

Answer:- b

For Q.7 – Q.10

Consider an instance of the 8 puzzle problem. We are given the state on the left side and we want to transform it into the state on the right. A problem with fewer restrictions on the actions is called a relaxed problem. The cost of an optimal solution to a relaxed problem is an admissible heuristic for the original problem. Consider two admissible heuristics Hamming Distance h1 and Manhattan Distance h2.

1	3	4
8		5
7	2	6

start

1	2	3
8		4
7	6	5

goal

7. Compute the Hamming Distance of the start state for the given instance h1(start).

Answer:- 5

8. Compute the Manhattan Distance of the start state for the given instance h2(start).

Answer:- 6

9. Find the number of moves required to reach the goal from the start using the Manhattan Distance heuristics.

Answer:- 6

10. How many states were required to be examined for reaching the goal using the Manhattan Distance heuristics?

Answer:- 5

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Assignment 2

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2019-08-21, 23:59 IST.

1) Heuristics created as solutions to relaxed problems are usually _____ because adding new actions should only reduce the solution cost and not increase it. ***1 point***

- A. Optimal
- B. Minimal cost
- C. Admissible
- D. Pessimistic

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. Admissible

2) Which of the following statement is false for inadmissible heuristics? ***1 point***

- A. Overestimate the cost.
- B. Trap good plans on the fringe.
- C. Breaks optimality.
- D. Underestimate the cost.

No, the answer is incorrect.
Score: 0

Accepted Answers:
D. Underestimate the cost.

3) A consistent heuristic is admissible; consistent heuristics places even _____ constraints on the heuristic. ***1 point***

- A. softer
- B. stricter
- C. weaker
- D. relaxed

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. stricter

4) The cooling strategy in Simulated Annealing does not determine the _____ ***1 point***

- A. maximum number of iterations in the search process.
- B. temperature-decrease steps.
- C. selection of a successor state.
- D. number of iterations for each step.

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. selection of a successor state.

5) Which of the following is not a property of the Hill-climbing algorithm? ***1 point***

- A. Terminates when a peak is reached.
- B. Does not look ahead of the immediate neighbors of the current state.
- C. Chooses randomly among the set of best successors.
- D. Does backtrack.

No, the answer is incorrect.
Score: 0

Accepted Answers:
D. Does backtrack.

6) A* generates an optimal solution under which of the following conditions? ***1 point***
Condition 1: If h(n) is an admissible heuristic and the search space is a tree.
Condition 2: If h(n) is a consistent heuristic and the search space is a graph.

- A. A* never generate an optimal solution.
- B. Condition 1 only
- C. Both Condition 1 and Condition 2
- D. Condition 2 only

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. Both Condition 1 and Condition 2

7) The _____ is a way of combining the advantages of both depth-first and breadth-first search into a single method. ***1 point***

- A. Iterative Deepening Depth First Search
- B. Bidirectional Search
- C. Best First Search.
- D. Depth Limited Search

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. Best First Search.

8) In heuristic search, the heuristic function, h(n), of a node n estimates the _____ ***1 point***

- A. cost to a goal from the current node n.
- B. cost from the start to the current node n.
- C. overall cost of the path through the node n.
- D. All of the above.

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. cost to a goal from the current node n.

9) Arc Consistency is a method of constraint propagation. The following is incorrect for arc consistency. ***1 point***

- A. Forward checking is substantially stronger than arc consistency.
- B. Detects failure early!
- C. Arc consistency must be applied repeatedly until no more inconsistencies remain.
- D. Provides a fast method of constraint propagation.

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Forward checking is substantially stronger than arc consistency.

10) An algorithm that is aware of the underlying constraint graph; determines where to jump back to, based on the actual conflict that it has recorded is _____. ***1 point***

- A. Chronological backtracking.
- B. Backjumping.
- C. Conflict-directed Backjumping.
- D. Backtracking.

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. Conflict-directed Backjumping.

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Assignment No. 3: Detailed Solution

10 Marks

Each question carries 01 Mark each.

Q1. Which ONE of the following statements is false w.r.t AND/OR Graphs?

A. A solution graph is analogous to a path in an ordinary graph.

B. Every OR node contributes exactly one selected successor to the solution graph.

C. Every AND node contributes all of its successors to the solution graph.

D. AO* always expands every node of the graph before terminating.

Correct Answer: (D)

Explanation: AO* is a best-first heuristic algorithm. It expands only the most promising partial solution graph and can terminate as soon as the best solution graph is found, leaving many nodes unexpanded.

Q2. Suppose an OR node has three successors. Assume the heuristic values represent the estimated remaining cost from each successor to a goal state.

Node	Edge cost	Heuristic
A	2	6
B	1	8
C	4	2

Which node is selected first by AO*?

A. Node A

B. Node B

C. Node C

D. Cannot be determined

Correct Answer: (C)

Explanation:

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```
f(A) = 2 + 6 = 8
f(B) = 1 + 8 = 9
f(C) = 4 + 2 = 6

AO* selects node with minimum f -> C (f = 6)
```

Q3. In game playing, which assumption distinguishes Minimax from ordinary shortest-path search?

A. The graph is cyclic.

B. The opponent always plays optimally.

C. Edge costs are negative.

D. The heuristic is admissible.

Correct Answer: (B)

Q4. A game such as Chess has a large branching factor and a very deep game tree.
Which of the following are correct?

I. Exhaustive search to terminal positions is impractical because the number of leaf nodes grows exponentially with search depth.
II. Limiting the search depth and applying a heuristic evaluation function at non-terminal nodes provides an estimate of the game state's utility.
III. Alpha-Beta pruning changes the Minimax decision obtained at the root by eliminating inferior branches.
IV. The final decision is based on backed-up utility estimates rather than exact game outcomes.

A. I and II only

B. I, II and IV only

C. II, III and IV only

D. All four statements

Correct Answer: (B)

Q5. Which of the following statements are true for searching an AND-OR graph?

I. To find solution in an AND-OR graph we need an algorithm similar to best-first search; but with the ability to handle the AND arc appropriately.
II. Algorithm should find a path from the start node to a set of nodes representing the solution states.

A. Both I and II
B. Neither I nor II
C. I Only
D. II Only

Correct Answer: (A)

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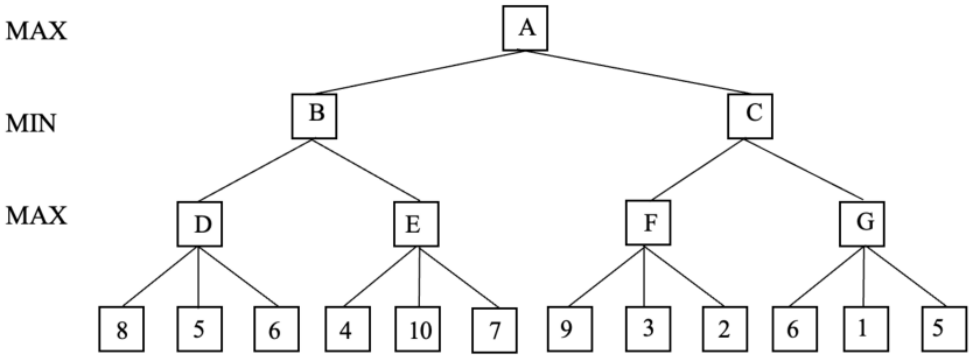
For Q6 – Q10

Consider the following game tree. The root node is a MAX node, and the levels alternate between MIN and MAX nodes. Terminal nodes contain the utility values shown. During Minimax and Alpha-Beta search, child nodes are evaluated from left to right.

Based on the game tree, answer Questions 6–10.



Based on the above data, answer the following questions.

Q6. After the complete evaluation of node B, what are the values of α and β at the root node A?

A. $\alpha = 6, \beta = +\infty$
B. $\alpha = 8, \beta = +\infty$
C. $\alpha = 10, \beta = +\infty$
D. $\alpha = 8, \beta = 8$

Correct Answer: (B)

Explanation: Minimax value of every node: D = 8, E = 10, B = 8, F = 9, G = 6, C = 6, A = 8.

Alpha – Beta root value = 8.

Leaves examined (in order): 8, 5, 6, 4, 10, 9, 3, 2, 6, 1, 5.

After evaluating B: at root A: alpha = 8, beta = infinity.

Q7. During left-to-right Alpha-Beta search, the first pruning occurs during the evaluation of which node?

A. Node D
B. Node E
C. Node F
D. Node G

Correct Answer: (B)

Explanation: First pruning happens at node E.

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Q8. Which one of the following leaf values is never examined during Alpha-Beta search?

A. 10
B. 7
C. 1
D. 5

Correct Answer: (B)

Explanation: Only the cut-off at E skips a leaf, and that leaf is the third child of E (value 7). Every other leaf is read.

Q9. During Alpha-Beta search, suppose node D has been evaluated but node E has not. Which of the following is true?

A. α at node B is 8.
B. β at node B is 8.
C. β at the root is 8.
D. α at the root is 8.

Correct Answer: (B)

Explanation: At node B, after child D and before E: alpha(B) = - infinity, beta(B) = 8.

Q10. Suppose the move ordering at every MAX node changes from best-first to worst-first. Which of the following is correct?

A. More nodes are likely to be evaluated.
B. Fewer nodes are guaranteed to be evaluated.
C. The Minimax value changes.
D. Alpha-Beta pruning becomes incorrect.

Correct Answer: (A)

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Fundamentals of Artificial Intelligence
MOOCs; July -Nov 2025
Assignment No. 3

10 Marks

Each question carries 01 Mark each.

Q1. The AO* algorithm can best be seen as the following TWO major operations:

A. 1. Top-down: graph-growing; 2. Bottom-up: cost-revising, SOLVE-labelling.
B. 1. Bottom-up: graph-growing; 2. Top-down: cost-revising, SOLVE-labelling.
C. 1. Top-down: graph-growing, SOLVE-labelling; 2. Bottom-up: cost-revising.
D. 1. Bottom-up: graph-growing, SOLVE-labelling; 2. Top-down: cost-revising.

Q2. For more complex games, such as chess or checker the AND/OR search to termination is out of question. Which of the following statements are true?

1. Our goal in searching such a game tree might be, instead, merely to find a good first move.
2. Extract from the search graph an estimate of the 'best' first move.

A. Both 1 and 2.
B. Only 2
C. Only 1
D. Both 1 and 2 are false.

Q3. Which of the following statements are true for searching an AND-OR graph?

I. To find solution in an AND-OR graph we need an algorithm similar to best-first search; but with the ability to handle the AND arc appropriately.
II. Algorithm should find a path from the start node to a set of nodes representing the solution states.

A. Both I and II
B. Neither I nor II
C. I Only
D. II Only

Q4. In the pay-off matrix of a zero-sum game, an outcome is _____ if the outcome is a minimum in its row and maximum in its column.

A. a critical point
B. an inflexion point
C. a saddle point
D. a stable point

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Q5. Assertion A Search procedure for Minimax that we have discussed is grossly inefficient strategy.

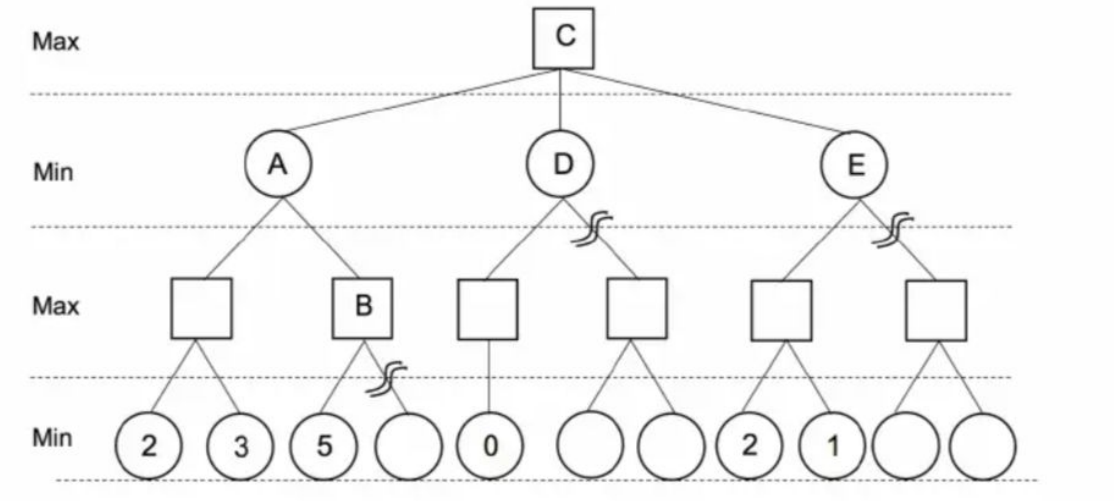
Reason R The search procedure separates completely the processes of search-tree generation and position evaluation.

Mark the correct choice as

A. Both A and R are true and R is the correct explanation for A
B. Both A and R are true but R is not the correct explanation for A
C. A is True but R is False
D. A is false but R is True

For Q6. – Q10.

Given below is a hypothetical minimax search space. Static values that shows the worth of leaf nodes are marked. Consider a left-to-right alpha-beta prune; pruning shown by pair of waved lines.



Q6. Start at C. Descent to full-ply depth. Back-up values as appropriate. The alpha value at node C after traversal of the leftmost branch is _____.

A. 5
B. 3
C. 0
D. 2

Q7. Applying alpha-beta pruning as appropriate, cut-off search below node B as shown is _____.

A. alpha pruning
B. beta pruning
C. incorrectly pruned
D. deep pruning

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Q8. Applying alpha-beta pruning as appropriate, cut-off search below node D as shown is _____.

A. alpha pruning
B. beta pruning
C. incorrectly pruned
D. deep pruning

Q9. Start at C. Descend to full-ply depth. Back up the values as appropriate. The beta value at node E is _____.

A. 5
B. 3
C. 0
D. 2

Q10. Descending to C's great grandchildren along the middle branch in a depth-first fashion, search below node D is pruned. This is because no matter what happens on its right branch the value of node D cannot be greater than _____.

A. 5
B. 3
C. 0
D. 2

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Assignment 3

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2019-08-21, 23:59 IST.

1) A zero-sum game is one where _____. ***1 point***

- A. The payoff is never zero.
- B. One player's gain is the other player's loss.
- C. No one wins.
- D. None of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. One player's gain is the other player's loss.

2) The minimax rule backs up values from the children of a node. For a MAX node, it backs up the _____. ***1 point***

- A. Maximum of the values of the children.
- B. Minimum of the values of the leaf node.
- C. Maximum of the values of the leaf node.
- D. Minimum of the values of the children.

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Maximum of the values of the children.

3) A solution graph of an AND-OR graph is analogous to a _____ in an ordinary graph. ***1 point***

- A. Path
- B. Cycle
- C. Sub-graph
- D. All of above.

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Path

4) The AO* algorithm can best be seen as the following TWO major operations: ***1 point***

- A. 1. Top-down: graph-growing; 2. Bottom-up: cost-revising, SOLVE-labelling.
- B. 1. Bottom-up: graph-growing; 2. Top-down: cost-revising, SOLVE-labelling.
- C. 1. Top-down: graph-growing, SOLVE-labelling; 2. Bottom-up: cost-revising.
- D. 1. Bottom-up: graph-growing, SOLVE-labelling; 2. Top-down: cost-revising.

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. 1. Top-down: graph-growing; 2. Bottom-up: cost-revising, SOLVE-labelling.

5) For a game tree: Each node has b children and a d-ply look ahead is performed. What is the number of leaf nodes to be examined? ***1 point***

- A. b^d
- B. b x d
- C. d^b
- D. b+d

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. b^d

6) Given a MIN node n; cutoff search below n, if beta(n) ≤ alpha(i) for some MAX node ancestor i of n, we have _____ cut-off. ***1 point***

- A. alpha
- B. beta
- C. deep
- D. min

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. alpha

7) Given a MAX node n; cutoff search below n, if alpha(n) ≥ beta(i) for some MIN node ancestor i of n, we have _____ cut-off. ***1 point***

- A. alpha
- B. beta
- C. deep
- D. min

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. beta

8) Alpha-beta is guaranteed to compute the same value for the root node as computed by _____ with less or equal computation. ***1 point***

- A. Breadth First Search
- B. Minimax
- C. Depth First Search
- D. None of the above.

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Minimax

9) For more complex games, such as chess or checker the AND/OR search to termination is out of question. Which of the following statements are true? ***1 point***

1. Our goal in searching such a game tree might be, instead, merely to find a good first move.
2. Extract from the search graph an estimate of the 'best' first move.

- A. Both 1 and 2.
- B. Only 2
- C. Only 1
- D. Both 1 and 2 are false.

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Both 1 and 2.

10) Using alpha-beta pruning, in the best case, you examine only (2b)^(d/2) leaf nodes. The result is you can search _____ as deep as minimax! ***1 point***

- A. Equal
- B. Twice
- C. Half
- D. One-fourth

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Twice

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Assignment No. 4: Detailed Solution

10 Marks

Each question carries 01 Mark each.

Q1. Which of the following statements correctly distinguishes Symbolic AI from Connectionist AI?

A. Symbolic AI is inspired primarily by the structure of the human brain.
B. Symbolic AI represents and manipulates knowledge using symbols, rules and structured representations.
C. Connectionist AI requires all knowledge to be represented explicitly as logical rules.
D. Symbolic AI cannot provide interpretable representations.

Correct Answer: (B)

Q2. Assertion A: A knowledge representation is fundamentally a surrogate.
Reason R: It acts as a substitute for the thing itself, enabling an agent to determine consequences by reasoning about the represented world.

Choose the correct answer:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, but R is not the correct explanation of A.
C. A is true, but R is false.
D. A is false, but R is true.

Correct Answer: (A)

Q3. Assertion A: Propositional logic has limited expressive power for representing structured knowledge.
Reason R: Propositional logic cannot directly represent properties of individuals, relations between individuals, or generalizations involving them.

Choose the correct answer:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, but R is not the correct explanation of A.
C. A is true, but R is false.
D. A is false, but R is true.

Correct Answer: (A)

Q4. Which of the following is logically equivalent to $\neg(P \rightarrow Q)$?

A. $\neg P \vee Q$
B. $P \wedge \neg Q$

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C. $\neg P \wedge Q$
D. $P \vee \neg Q$

Correct Answer: (B)

Q5. Which statement correctly distinguishes a *predicate* from a *function* in First-Order Logic?

A. Both always evaluate to truth values.
B. A predicate evaluates to an object, whereas a function evaluates to a truth value.
C. A predicate applied to arguments produces a proposition, whereas a function applied to arguments denotes an object.
D. Functions can have arguments, but predicates cannot.
Correct Answer: (C)

For Q6- Q10

Consider the following predicates:

- $\text{Robot}(x)$: x is a robot.
- $\text{Autonomous}(x)$: x is autonomous.
- $\text{Mobile}(x)$: x is mobile.

Q6. Which expression represents: All robots are autonomous.

A. $\forall x\big(\text{Robot}(x) \wedge \text{Autonomous}(x)\big)$
B. $\forall x\big(\text{Robot}(x) \rightarrow \text{Autonomous}(x)\big)$
C. $\exists x\big(\text{Robot}(x) \rightarrow \text{Autonomous}(x)\big)$
D. $\exists x\big(\text{Robot}(x) \wedge \text{Autonomous}(x)\big)$
Correct Answer: (B)

Q7. Which expression represents: Some robots are mobile.

A. $\forall x\big(\text{Robot}(x) \rightarrow \text{Mobile}(x)\big)$
B. $\exists x\big(\text{Robot}(x) \wedge \text{Mobile}(x)\big)$
C. $\forall x\big(\text{Robot}(x) \wedge \text{Mobile}(x)\big)$
D. $\neg\exists x\big(\text{Robot}(x) \wedge \text{Mobile}(x)\big)$
Correct Answer: (B)

Q8. Which expression represents: No robot is mobile.

A. $\forall x\big(\text{Robot}(x) \rightarrow \text{Mobile}(x)\big)$
B. $\exists x\big(\text{Robot}(x) \wedge \neg\, \text{Mobile}(x)\big)$
C. $\neg\exists x\big(\text{Robot}(x) \wedge \text{Mobile}(x)\big)$
D. $\exists x\big(\neg\, \text{Robot}(x) \wedge \text{Mobile}(x)\big)$
Correct Answer: (C)

Q9. The negation of $\forall x\big(\text{Robot}(x) \rightarrow \text{Autonomous}(x)\big)$ is logically equivalent to:

A. $\forall x\big(\text{Robot}(x) \wedge \neg\, \text{Autonomous}(x)\big)$
B. $\exists x\big(\text{Robot}(x) \wedge \neg\, \text{Autonomous}(x)\big)$

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C. $\exists x(\neg \text{Robot}(x) \wedge \text{Autonomous}(x))$

D. $\forall x(\text{Robot}(x) \rightarrow \neg\, \text{Autonomous}(x))$

Correct Answer: (B)

Q10. Suppose $\forall x(\text{Robot}(x) \rightarrow \text{Autonomous}(x))$ and $\forall x(\text{Autonomous}(x) \rightarrow \text{Mobile}(x))$ are true. Which conclusion necessarily follows?

A. $\forall x(\text{Robot}(x) \rightarrow \text{Mobile}(x))$

B. $\forall x(\text{Mobile}(x) \rightarrow \text{Robot}(x))$

C. $\exists x(\text{Robot}(x) \wedge \text{Mobile}(x))$

D. $\forall x(\text{Robot}(x) \rightarrow \neg\text{Mobile}(x))$

Correct Answer: (A)

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Assignment Number 4: Detailed Solution
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Each question carries 01 Mark each.

Q1. Identity the correct combination of terms and definitions below.

Terms	Definition
P. data	X. primitive verifiable facts, of any representation.
Q. information	Y. relation among sets of data, that is very often used for further information deduction.
R. knowledge	Z. interpreted data

A. P:X; Q:Z; R:Y

B. P:Y; Q:Z; R:X

C. P:X; Q:Y; R:Z

D. P:Y; Q:Z; R:X

Q2. A physical symbol system has the necessary and sufficient means for general intelligent action. Identify the correct statements with regards to Symbol System Hypothesis

A. Knowledge may be represented as symbol structures.

B. Represents a computational system inspired by the human brain.

C. Intelligent behaviour can be achieved through manipulation of symbol structures.

D. Human thinking is a kind of symbol manipulation.

Q3. Consider which of the following statements are correct w.r.t. entailment in first-order logic being semi-decidable.

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Assignment Number 4: Detailed Solution
Indian Institute of Technology Guwahati

I. Algorithms exist that say yes to every entailed sentence

II. No algorithm exists that says no to every non-entailed sentence.

A. Both I and II

B. Either I or II

C. I only

D. II only

Q4. Entailment for FOL is _____: algorithms exist that say yes to every entailed sentence, but no algorithm exists that also says no to every non-entailed sentence.

A. Decidable

B. Semidecidable

C. Efficient

D. Undecidable

Q5. Consider the statement $\neg(P \wedge Q) \equiv (\neg P \vee \neg Q)$. Identify which of the following are correct.

A. Formulas $\neg(P \wedge Q)$ and $(\neg P \vee \neg Q)$ have the same truth tables.

B. The symbol $\equiv$ is a connective.

C. The statement is a propositional formula which evaluates to a truth value.

D. The statement means `the two formulas are equivalent.'

Q6. The negation of a universal statement $(\forall x\ \varphi)$ is logically equivalent to the following existential statement

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NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
Assignment Number 4: Detailed Solution
Indian Institute of Technology Guwahati

A. $\exists x \neg \varphi$

B. $\neg \exists x \varphi$

C. $\forall x \neg \varphi$

D. $\neg \forall x \varphi$

Q7. Assertion A Propositional Logic is a weak Language.

Reason R Every statement in propositional logic consists of propositional variables combined via propositional connectives.

Mark the correct choice as

A. Both A and R are true and R is the correct explanation for A
B. Both A and R are true but R is not the correct explanation for A.
C. A is True but R is False
D. A is false but R is True

Q8. Assertion A A knowledge representation is fundamentally a surrogate.

Reason R A knowledge representation is a substitute for the thing itself; used to enable an entity to determine consequences by reasoning about the world.

Mark the correct choice as

A. Both A and R are true and R is the correct explanation for A
B. Both A and R are true but R is not the correct explanation for A
C. A is True but R is False
D. A is false but R is True

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Fundamentals of Artificial Intelligence
Assignment Number 4: Detailed Solution
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For Q9 – Q10

Consider the predicates listed below:

Professor(x)	x is a Professor.
Person(x)	x is a person.
Dean(x)	x is a Dean.
Friend(x,y)	x is a friend of y.
Know(x,y)	x knows y.
Criticize(x,y)	x criticizes y.

Q9. Translate the following English statement into First Order Logic statement.

All professors consider the dean a friend or do not know him.

A. $\forall x \left(\exists y \left(\text{Professor}(x) \wedge \text{Dean}(y) \rightarrow \text{Friend}(x,y) \vee \neg\ \text{Know}(x,y)\right)\right)$

B. $\forall x \left(\forall y \left(\text{Professor}(x) \wedge \text{Dean}(y) \rightarrow \text{Friend}(x,y) \vee \neg\ \text{Know}(x,y)\right)\right)$

C. $\forall x \left(\exists y \left(\text{Professor}(x) \wedge \text{Dean}(y) \wedge\ \text{Friend}(x,y) \vee \neg\ \text{Know}(x,y)\right)\right)$

D. $\forall x \left(\forall y \left(\text{Professor}(x) \wedge \text{Dean}(y) \leftrightarrow\ \text{Friend}(x,y) \vee \neg\ \text{Know}(x,y)\right)\right)$

Q10. Translate the following English statement into First Order Logic statement.

Person only criticize person that are not their friends.

A. $\forall x \left(\forall y \left(\text{Person}(x) \wedge \text{Person}(y) \wedge \text{Criticize}(x,y) \rightarrow \neg\ \text{Friend}(y,x)\right)\right)$

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NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
Assignment Number 4: Detailed Solution
Indian Institute of Technology Guwahati

B. $\forall x \left(\forall y \left(\text{Person}(x) \land \text{Person}(y) \rightarrow \text{Criticize}(x,y) \land \neg\ \text{Friend}(y,x)\right)\right)$

C. $\forall x \left(\forall y \left(\text{Criticize}(x,y) \rightarrow \neg\ \text{Friend}(y,x)\right)\right)$

D. $\forall x \left(\forall y \left(\neg\ \text{Friend}(x,y) \rightarrow \text{Criticize}(y,x)\right)\right)$

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Assignment 4

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2019-08-28, 23:59 IST.

1) Pragmatics of a knowledge representation language specify *1 point*

- A. which group of symbols, arranged in what way, are to be considered properly formed.
- B. what the well-formed expressions are supposed to mean.
- C. when are the well-formed expressions useful.
- D. how the meaningful expressions are to be used.

No, the answer is incorrect.
Score: 0

Accepted Answers:
D. how the meaningful expressions are to be used.

2) Consider which of the following statements are correct w.r.t. entailment in first-order logic being semi-decidable. *1 point*

I. Algorithms exist that say yes to every entailed sentence
II. No algorithm exists that says no to every non-entailed sentence.

- A. Both I and II
- B. Either I or II
- C. I only
- D. II only

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Both I and II

3) First order logic (FOL) does not allow quantification over predicate symbols or function symbols; therefore, FOL is called _____. *1 point*

- A. Predicate logic.
- B. First-order.
- C. Sentential logic.
- D. Predicate Calculus.

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. First-order.

4) In First order logic, if a set of sentence is _____ then resolution will always be able to derive a contradiction. *1 point*

- A. Unsatisfiable
- B. Conjunction of literals
- C. Disjunction of literals
- D. Satisfiable

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Unsatisfiable

5) If a sentence has no free variables, it is called a _____ sentence; If it has neither free nor bound variables, it is called a _____ sentence. *1 point*

- A. Closed; ground
- B. Ground; closed
- C. Formula; closed
- D. Closed; formula

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Closed; ground

6) The _____ for a predicate variable is the set of values that may be assigned to the variable. *1 point*

- A. universe of discourse
- B. interpretation
- C. truth value
- D. evaluation

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. universe of discourse

7) For propositional logic, which of the following statements are true. *1 point*

I. Hard to identify “individuals”.
II. Can’t directly talk about properties of individuals or relations between individuals.
III. Generalizations, patterns, regularities can’t easily be represented.

- A. Only I
- B. Only II
- C. I and II
- D. I, II and III

No, the answer is incorrect.
Score: 0

Accepted Answers:
D. I, II and III

8) A knowledge representation is fundamentally a _____; a substitute for the thing itself used to enable an entity to determine consequences by reasoning about the world. *1 point*

- A. reasoning schema
- B. surrogate
- C. constraint
- D. language

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. surrogate

9) The _____ states that: Knowledge may be represented as symbol structures; Intelligent behaviour can be achieved through manipulation of symbol structures. *1 point*

- A. Symbol System Hypothesis
- B. Physical System Hypothesis
- C. Physical Symbol Hypothesis
- D. Representation Hypothesis

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Symbol System Hypothesis

10) In FOL, a _____ is a property that a variable or a finite collection of variables can have. *1 point*

- A. function
- B. relation
- C. predicate
- D. constant

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. predicate

NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
MOOCs; July - Dec 2026
Indian Institute of Technology Guwahati
Assignment No. 5: Detailed Solution

10 Marks

Each question carries 01 Mark each.

Q1. Which ONE of the following best describes Declarative Semantics in First Order Logic?

A. It specifies the order in which inference rules are applied.
B. It associates symbols in the language with objects, functions and relations of a conceptualization to determine the truth of sentences.
C. It converts quantified sentences into propositional logic.
D. It defines the syntax of well-formed formulae.
Correct Answer: (B)

Q2. Consider the following statements regarding conceptualization:

I. Every conceptualization contains a universe of discourse.
II. Every conceptualization specifies concepts and relationships among the objects in the domain.

Choose the correct option.

A. I only
B. II only
C. Both I and II
D. Neither I nor II
Correct Answer: (C)

Q3. Facts and rules that attempt to capture the important facts and concepts about a domain are collectively referred to as:

A. Interpretations
B. Axioms
C. Models
D. Proofs
Correct Answer: (B)

Q4. Which inference rule is NOT a sound deductive rule in classical first-order logic?

A. Modus Ponens
B. Modus Tollens
C. Resolution
D. Abduction
Correct Answer: (D)

Indian Institute of Technology Guwahati

Q5. Which inference step is illustrated below?

$\forall x$ Likes(x, IceCream). Therefore Likes(Rahul, IceCream).

A. Existential Generalization
B. Universal Instantiation
C. Existential Instantiation
D. Resolution

Correct Answer: (B)

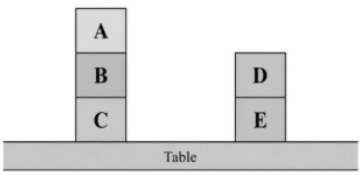
For Q6 – Q10

Consider the following Blocks World:
The predicates have the following meanings:

- on(x, y) : Block x is immediately on top of block y.
- above(x, y) : Block x is somewhere above block y (not necessarily immediately).
- clear(x) : No block is on top of block x.
- table(x) : Block x is directly on the table.

The knowledge base consists of the following facts:

- on(A, B)
- on(B, C)
- on(D, E)
- table(C)
- table(E)
- clear(A)
- clear(D)



The following rules are also known to be true.

- $\forall x \forall y\ (on(x, y) \rightarrow above(x, y))$
- $\forall x \forall y \forall z\ ((above(x, y) \wedge above(y, z)) \rightarrow above(x, z))$
- $\forall x\ (clear(x) \rightarrow \neg \exists y\ on(y, x))$

Q6. Which fact can be inferred using a single application of Modus Ponens?

A. above(A, C)
B. above(A, B)
C. clear(C)
D. table(A)

Correct Answer: (B)

Indian Institute of Technology Guwahati

Q7. Which relation requires applying the transitivity rule?

A. above(A, B)
B. above(B, C)
C. above(A, C)
D. table(C)
Correct Answer: (C)

Q8. If the fact on(B,C) is removed, which inferred relation can no longer be proved?

A. above(D, E)
B. above(A, B)
C. above(A, C)
D. clear(A)
Correct Answer: (C)

Q9. Which statement is NOT entailed by the given knowledge base?

A. above(D, E)
B. $\neg\exists$y on(y, A)
C. $\neg\exists$y above(y, A)
D. above(A, C)
Correct Answer: (C)

Q10. Which of the following propositions is logically entailed by the knowledge base but is neither explicitly given as a fact nor derivable using a single application of Modus Ponens?

A. above(A, B)
B. above(D, E)
C. $\neg\exists$y on(y, A)
D. above(A, C)

Correct Answer: (D)

Indian Institute of Technology Guwahati

NPTEL Online Certification Course

Fundamentals of Artificial Intelligence

Assignment Number 05: Detailed Solution

Indian Institute of Technology Guwahati

Each question carries 01 Mark each.

Q1. In First Order Logic, the elements of a conceptualization include _____

A. universe of discourse and an interpretation.

B. well-formed expression.

C. objects, relations and functions.

D. meaningful expression.

Q2. P, Q and R are logical propositions. Identify the statements as tautology, and / or contradiction?

I. $\big((P \vee Q) \wedge R\big) \leftrightarrow ((P \wedge R) \vee (Q \wedge R))$

II. $(P \leftrightarrow Q) \wedge (Q \leftrightarrow R) \wedge \neg(P \leftrightarrow R)$

A. I. Tautology; II. Contradiction.

B. I. Contradiction; II. Tautology.

C. Both Tautologies.

D. Both Contradictions.

Q3. Facts and rules that attempt to capture all of the (important) facts and concepts about a domain are referred to as _____

A. definitions.

B. axioms.

C. theorems.

D. truth values.

Q4. Skolemization is a procedure for systematic elimination of the _____ in a first-order formula in a prenex form, by introducing new constant and functional symbols.

Indian Institute of Technology Guwahati

NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
Assignment Number 05: Detailed Solution
Indian Institute of Technology Guwahati

A. existential quantifiers

B. universal quantifiers

C. variable

D function

Q5. An axiom is a well-formed formula that is asserted to be true without proof. In an AI system, the axioms would be:

A. The domain-specific knowledge rules in the database.

B. The control strategy rules selected by the inference mechanism.

C. The input data supplied by the user.

D. The output data obtained through the inference process.

Q6. A sentence in a formal language is true if and only if it accurately describes the world according to our conceptualization. _____ is a mapping between elements of the language and elements of a conceptualization.

A. Declarative semantics.

B. An interpretation.

C. A representation.

D. An assignment.

Q7. **Assertion A**: Any predicate calculus well-formed formula can be converted to a set of clauses.

Reason R: The *prenex* form consists of a string of quantifiers called *prefix* followed by a quantifier-free formula called the *matrix*.

A. Both A and R are true and R is the correct explanation for A

B. Both A and R are true but R is not the correct explanation for A

C. A is True but R is False

D. A is false but R is True

Indian Institute of Technology Guwahati

NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
Assignment Number 05: Detailed Solution
Indian Institute of Technology Guwahati

Q8. Consider the predicate Likes(x,y): x likes y.

Everyone likes ice cream. Is it possible to convey the same meaning using an existential statement? If yes, give the existential statement.

A. No.

B. Yes. $\forall x\ \text{Likes}(x, \text{Icecream})$

C. Yes. $\exists x\ \text{Likes}(x, \text{Icecream})$

D. Yes. $\neg\ \exists x \neg \text{Likes}(x, \text{Icecream})$

Q9. Write the FOL formula obtained by introducing the skolem function Support(x) in the NEXT step towards arriving at the Clausal Normal Form for the following

$\forall x[\neg \text{Brick}(x) \vee (\exists y[\text{On}(x,y) \wedge \neg \text{Pyramid}(y)] \wedge \forall y[\neg \text{On}(x,y) \vee \neg \text{On}(y,x)])]$

A. $\forall x[\neg \text{Brick}(x) \vee ([\text{On}(x, \text{Support}(x)) \wedge \neg \text{Pyramid}(y)] \wedge \forall y[\neg \text{On}(x,y) \vee \neg \text{On}(y,x)])]$

B. $\forall x[\neg \text{Brick}(x) \vee (\exists y[\text{On}(x, \text{Support}(x)) \wedge \neg \text{Pyramid}(\text{Support}(x))] \wedge \forall y[\neg \text{On}(x,y) \vee \neg \text{On}(y,x)])]$

C. $\forall x[\neg \text{Brick}(x) \vee ([\text{On}(x, \text{Support}(x)) \wedge \neg \text{Pyramid}(\text{Support}(x))] \wedge \forall y[\neg \text{On}(x,y) \vee \neg \text{On}(y,x)])]$

D. $\forall x[\neg \text{Brick}(x) \vee ([\text{On}(x, \text{Support}(x)) \wedge \neg \text{Pyramid}(\text{Support}(x))] \wedge [\neg \text{On}(x, \text{Support}(x)) \vee \neg \text{On}(\text{Support}(x), x)])]$

Q10. A definition of a predicate is a *biconditional*, and can be decomposed into a necessary and sufficient descriptions. Which of the following statement are true with regards to definition of `Brother'

I. Being a `Male' is a necessary condition for being a `Brother', but it is not sufficient.

II. Being a `Male' `Sibling' is a necessary and sufficient condition for being a `Brother'.

A. I Only

B. II Only

C. Both I and II

D. None

Indian Institute of Technology Guwahati

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Assignment 5

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2019-09-04, 23:59 IST.

1) In First Order Logic, an interpretation I is a mapping between elements of the language and elements of a _____ ***1 point***

- A. representation.
- B. well-formed expression.
- C. conceptualization.
- D. meaningful expression.

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. conceptualization.

2) Consider which of the following statements are correct w.r.t. satisfiability of logical sentences based on the logical operators involved. ***1 point***

I. Universally quantified sentence is satisfied if and only if the enclosed statement is satisfied for all assignments of the quantified variable.
II. Existentially quantified sentence is satisfied if and only if the enclosed statement is satisfied for few assignments of the quantified variable.

- A. Both I and II
- B. Either I or II
- C. I only
- D. II only

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. I only

3) Facts and rules that attempt to capture all of the (important) facts and concepts about a domain are referred to as _____ ***1 point***

- A. Predicate logic.
- B. Axioms.
- C. Theorems.
- D. Truth values.

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Axioms.

4) Also known as skolemization; the _____ is a convenient way to reason about the unknown object, rather than the existential quantifier. ***1 point***

- A. Skolem constant
- B. Skolem variable
- C. Universal variable
- D. Existential variable

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Skolem constant

5) A clause defined as a well-formed formula consisting of a _____ of literals. ***1 point***

- A. disjunction
- B. set
- C. conjunction
- D. array

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. disjunction

6) _____ helps immensely in search as it dramatically reduces the number of resolvents that can be inferred from two input clauses. ***1 point***

- A. Conjunctive Normal Form
- B. Most General Unifiers
- C. Resolution
- D. Breadth-first Strategy

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Most General Unifiers

7) Given loyal(x, y): x is loyal to y; The logic statement ∀"x" ∃y loyal(x, y) refers to ***1 point***

- A. Everyone is loyal to everyone.
- B. Everyone is loyal to someone.
- C. Someone is loyal to everyone.
- D. Someone is loyal to someone.

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Everyone is loyal to someone.

8) An interpretation of a set of FOL sentences, such that every sentence is true, is a _____. ***1 point***

- A. Model
- B. Tautology
- C. Formalization
- D. Proof

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Model

9) In First order logic, if a set of sentence is _____ then resolution will always be able to derive a contradiction. ***1 point***

- A. Unsatisfiable
- B. Conjunction of literals
- C. Disjunction of literals
- D. Satisfiable

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Unsatisfiable

10) Given loves(x, y): x loves y; The logic statement ∃y∀"x" loves(x, y) refers to ***1 point***

- A. Everyone loves everyone.
- B. Everyone loves someone.
- C. Someone loves everyone.
- D. Someone loves someone

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. Someone loves everyone.

NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
MOOCs; July - Dec 2026
Indian Institute of Technology Guwahati
Assignment No. 6: Detailed Solution

10 Marks

Each question carries 01 Mark each.

Q1. Which of the following statements are correct regarding entailment in First-Order Logic?

I. There exists an algorithm that halts and answers YES whenever $KB \models \alpha$.
II. There is no algorithm that always halts and answers NO whenever $KB \models/\alpha$.

Choose the correct option.
A. Both I and II
B. I only
C. II only
D. Neither I nor II
Correct option: (A)

Q2. In first order logic, if a set of sentences is _____ then resolution will always be able to derive a contradiction.
A. unsatisfiable
B. conjunction of literals
C. disjunction of literals
D. satisfiable

Correct option: (A)

Q3. A clause C_1 subsumes a clause C_2 if there exists a substitution s such that
A. $C_1 s \subseteq C_2$
B. $C_2 s \subseteq C_1$
C. $C_1 \subseteq C_2 s$
D. $C_1 s = C_2$
Correct option: (A)

Q4. Which inference step is illustrated below?

$\forall x$ Likes(x, IceCream) Therefore Likes(Rahul, IceCream)

A. Existential Generalization
B. Universal Instantiation

Indian Institute of Technology Guwahati

C. Existential Instantiation
D. Resolution

Correct option: (B)

Q5. The _____ strategy restricts each resolution step so that one parent clause must be either a member of the original clause set, or an ancestor of the other parent clause in the derivation.

A. Unit resolution
B. Input resolution
C. Linear resolution
D. Set-of-support

Correct option: (C)

From Q6 to Q10

Consider the following representation in First Order Logic:

1. A horse that is registered for today's race is not a thoroughbred.
2. Every horse registered for today's race has won a race this year.
3. A horse that has won a race this year is not a thoroughbred.

P(x): x is registered for today's race.
Q(x): x is a thoroughbred.
R(x): x has won a race this year.
U = all horses

Q6. In first-order logic, each variable refers to some object in a set called the _____.
A. domain of discourse
B. knowledge base
C. interpretation
D. semantics
Correct option: (A)

Q7. Translate the following English sentence to FOL.

Every horse that is registered for today's race is not a thoroughbred.

A. $\forall x\,[P(x) \vee Q(x)]$
B. $\forall x[P(x) \rightarrow \neg Q(x)]$
C. $\exists x[P(x) \wedge \neg Q(x)]$
D. $\forall x\,[P(x) \rightarrow Q(x)]$
Correct option: (B)

Q8. Translate the following English sentence to First Order Logic.

Every horse registered for today's race has won a race this year.

Indian Institute of Technology Guwahati

A. $\exists x[P(x) \wedge R(x)]$
B. $\forall x\,[P(x) \rightarrow R(x)]$
C. $\exists x[P(x) \wedge \neg R(x)]$
D. $\forall x\,R(x)$
Correct option: (B)

Q9. Translate the following English sentence to FOL.

Every horse that has won a race this year is not a thoroughbred.

A. $\forall x\,[R(x) \vee Q(x)]$
B. $\forall x[R(x) \rightarrow \neg Q(x)]$
C. $\exists x[R(x) \wedge \neg Q(x)]$
D. $\forall x\,[R(x) \rightarrow Q(x)]$
Correct option: (B)

Q10. P(c) for some element c
$\exists x\,P(x)$

We will be using this to prove statements of the form $\exists x\,P(x)$. This is _____.

A. Universal generalization
B. Universal Instantiation
C. Existential generalization
D. Existential instantiation
Correct option: (C)

Indian Institute of Technology Guwahati

NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
Assignment Number 06: Detailed Solution
Indian Institute of Technology Guwahati

Each question carries 01 Mark each.

Q1. Consider which of the following statements are correct w.r.t. entailment in first-order logic being semi-decidable.

I. Algorithms exist that say yes to every entailed sentence

II. No algorithm exists that says no to every non-entailed sentence.

A. Both Statement I and II.

B. Either Statement I or II.

C. Statement I only.

D. Statement II only

Q2. Under which of the following conditions, a clause $\{L_i\}$ subsumes a clause $\{M_i\}$, if there exists a substitution 's'?

Condition 1: $\{L_i\}$s is a subset of $\{M_i\}$

Condition 2: $\{M_i\}$s is a subset of $\{L_i\}$

A. No clause ever subsumes another clause.
B. Condition 1 only
C. Both Condition 1 and Condition 2
D. Condition 2 only

Q3. Let p, q, r, and s represent the following propositions

p: $x \in \{8,9,10,11,12\}$.

q: x is a composite number.

r: x is a perfect square.

s: x is a prime number.

The integer $x \geq 2$ which satisfies $\neg ((p \Rightarrow q) \wedge (\neg r \vee \neg s))$ is:

Indian Institute of Technology Guwahati

NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
Assignment Number 06: Detailed Solution
Indian Institute of Technology Guwahati

A. 8

B. 9

C. 11

D. 12

Q4. Consider the first-order logic sentence

F: ∀ x (∃ y R(x,y)).

Assuming non-empty logical domains, which of the sentences below are implied by F?

I. ∃y (∃x R(x,y))

II. ∃y (∀x R(x,y))

III. ∀y (∃x R(x,y))

IV. ¬∃x (∀y ¬ R(x,y))

A. IV only

B. I and IV only

C. II only

D. II and III only

Q5. Answer extraction involves converting a *refutation tree* to a *proof tree* with statement at the root that can be used as an answer. Which of the following are true w.r.t the above process of answer extraction?

I. Convert every clause arising from the negation of the goal well-formed formula into a tautology.

II. The statement at the root of the *modified proof tree* logically follows from the axioms and the tautologies.

A. Either I or II

B. Both I and II

Indian Institute of Technology Guwahati

NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
Assignment Number 06: Detailed Solution
Indian Institute of Technology Guwahati

C. I only

D. II only

For Q6 – Q10

Consider the following representation in First Order Logic:

Jack owns a dog.

Jill owns a cat.

Every dog owner is an animal lover.

No animal lover kills an animal.

Either Jack or Curiosity killed the cat named Tuna.

Use the predicates listed below:

Dog(x)	x is a dog.
Cat(x)	x is a cat.
Animal(x)	x is an animal.
AnimalLover(x)	x is an animal lover.
Kills(x,y)	x kills y.
Owns(x,y)	x owns y.

Q6. In first-order logic, each variable refers to some object in a set called the _____.

A. domain of discourse
B. knowledge base

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NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
Assignment Number 06: Detailed Solution
Indian Institute of Technology Guwahati

C. interpretation
D. semantics

Q7. Translate the following English sentence to FOL.

Jack owns a dog.

A. $\exists x\ (Dog(x) \rightarrow Owns(Jack, x))$
B. $\forall x\ (Dog(x) \wedge Owns(Jack, x))$
C. $\exists x\ (Dog(x) \wedge Owns(Jack, x))$
D. $\forall x\ (Dog(x) \rightarrow Owns(Jack, x))$

Q8. Translate the following English sentence to FOL.

Every dog owner is an animal lover.

A. $\forall x\ (\forall y\ Dog(y) \wedge Owns(x,y)) \rightarrow AnimalLover(x)$
B. $\forall x\ (\exists y\ Dog(y) \wedge Owns(x,y)) \rightarrow AnimalLover(x)$
C. $\forall y\ \exists x\ (Dog(x) \wedge Owns(x,y)) \wedge AnimalLover(x)$
D. $\forall y\ (\exists x\ Dog(y) \wedge Owns(x,y)) \rightarrow AnimalLover(x)$

Q9. Translate the following English sentence to FOL.

No animal lover kills an animal.

A. $\forall x\ AnimalLover(x) \rightarrow (\forall y\ Animal(y) \rightarrow \neg\ Kills(x,y))$

Indian Institute of Technology Guwahati

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NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
Assignment Number 06: Detailed Solution
Indian Institute of Technology Guwahati

B. $\forall x \exists y$ (Animal(y) $\wedge$ AnimalLover(x) $\wedge \neg$ Kills(x,y))
C. $\exists x \exists y$ (Animal(x) $\wedge$ AnimalLover(y) $\wedge \neg$ Kills(x,y))
D. $\forall x \forall y$ (AnimalLover(x) $\wedge$ Animal(y)) $\rightarrow \neg$ Kills(x,y)

Q10. Who killed the cat named Tuna?

A. Curiosity
B. Jack
C. Tony
D. Jill

Indian Institute of Technology Guwahati

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Assignment 6

Assignment not submitted **Due date: 2024-09-04, 23:59 IST.**

1) Under which of the following conditions, a clause {Li} subsumes a clause {Mi}, if there exists a substitution `s'? ***1 point***

Condition 1: {Li}s is a subset of {Mi}

Condition 2: {Mi}s is a subset of {Li}

- ○ A. No clause ever subsumes another clause.
- ○ B. Condition 1 only
- ○ C. Both Condition 1 and Condition 2
- ◉ D. Condition 2 only

2) Rule of inference, called _____ when added to resolution principle, guarantees refutation completeness, even when involving equality. ***1 point***

- ○ A. Demodulation
- ◉ B. Paramodulation
- ○ C. Modulation
- ○ D. Equimodulation

For Q3 to Q6

Consider the following representation in First Order Logic:

Animal(x)	: x is an animal.
Dog(x)	: x is a dog.
AnimalLover(x)	: x is an animal lover.
Kills(x,y)	: x kills y.
Owns(x,y)	: x owns y.

3) In first-order logic, each variable refers to some object in a set called the _____. ***1 point***

- (●) A. domain of discourse
- ( ) B. knowledge base
- ( ) C. interpretation
- ( ) D. semantics

4) Translate the following English sentence to First Order Logic statement. ***1 point***

Jack owns a dog.

- ( ) A. $\exists x(Dog(x) \rightarrow Owns(Jack,x))$
- ( ) B. $\forall x(Dog (x) \wedge Owns(Jack,x))$
- (●) C. $\exists x(Dog(x) \wedge Owns(Jack,x))$
- ( ) D. $\forall x(Dog (x) \rightarrow Owns(Jack,x))$

5) Translate the following English sentence to First Order Logic statement. ***1 point***

Every dog owner is an animal lover.

- ( ) A. $\forall x (\forall y\ Dog (y) \wedge Owns(x,y)) \rightarrow AnimalLover(x)$
- (●) B. $\forall x (\exists y\ Dog (y) \wedge Owns(x,y)) \rightarrow AnimalLover(x)$
- ( ) C. $\forall x \exists x (Dog (x) \wedge Owns(x,y)) \wedge AnimalLover(x))$
- ( ) D. $\forall y (\exists x\ Dog (y) \wedge Owns(x,y)) \rightarrow AnimalLover(x)$

6) Translate the following English sentence to First Order Logic statement. ***1 point***

No animal lover kills an animal.

- (●) A. $\forall x\ AnimalLover(x) \rightarrow (\forall y\ Animal (y) \rightarrow \neg\ Kills(x,y))$
- ( ) B. $\forall x \exists y\ (Animal (y) \wedge AnimalLover(x) \wedge \neg\ Kills(x,y))$
- ( ) C. $\exists x\ \exists y\ (Animal (x) \wedge AnimalLover (y) \wedge \neg Kills(x,y))$
- ( ) D. $\forall x\ \forall y\ (AnimalLover(x) \wedge Animal (y)) \rightarrow \neg\ Kills(x,y)$

For Q7 to Q10

Consider the following representation in First Order Logic:

P(x): x is registered for today's race.
Q(x): x is a thoroughbred.
R(x): x has won a race this year.
U = all horses

Axioms: A1. A horse that is registered for today's race is not a thoroughbred.
A2. Every horse registered for today's race has won a race this year.

Theorem: Th1. Therefore a horse that has won a race this year is not a thoroughbred.

7) Translate the following English sentence to FOL. ***1 point***

A horse that is registered for today's race is not a thoroughbred.

- A. $\forall x\,[P(x) \vee Q(x)]$
- B. $\forall x[P(x) \rightarrow \neg\, Q(x)]$
- C. $\exists x[P(x) \wedge \neg\, Q(x)]$
- D. $\forall x\,[P(x) \rightarrow Q(x)]$

8) Translate the following English sentence to First Order Logic. ***1 point***

Every horse registered for today's race has won a race this year.

- A. $\exists x[P(x) \wedge R(x)]$
- B. $\forall x\,[P(x) \rightarrow R(x)]$
- C. $\exists x[P(x) \wedge \square R(x)]$
- D. $\forall x\,R(x)$

9) Translate the following English sentence to FOL. ***1 point***

A horse that has won a race this year is not a thoroughbred.

- A. $\forall x\,[R(x) \vee Q(x)]$
- B. $\forall x[R(x) \rightarrow \neg\, Q(x)]$
- C. $\exists x[R(x) \wedge \neg\, Q(x)]$
- D. $\forall x\,[R(x) \rightarrow Q(x)]$

10) $P(c)$ for some element c ***1 point***

$\exists x\,P(x)$

We will be using this to prove statements of the form $\exists x\,P(x)$. This is _____.

- A. Universal generalization
- B. Universal Instantiation
- C. Existential generalization
- D. Existential instantiation

You may submit any number of times before the due date. The final submission will be considered for grading.

Submit Answers

Assignment 6

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2019-09-11, 23:59 IST.

1) First order logic (FOL) is called `first order' because *1 point*

- A. FOL does not allow quantification over predicate symbols or function symbols.
- B. Predicates are first-order relations.
- C. FOL comes first in the logics involving quantifiers.
- D. Propositional logic is zero-order

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. FOL does not allow quantification over predicate symbols or function symbols.

2) The process of removing detail from a given state representation is called_____ *1 point*

- A. Extraction
- B. Abstraction
- C. Information Retrieval
- D. Mining

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Abstraction

3) First-order resolution requires the clause to be in Conjunctive Normal Form, which are *1 point*

- A. Disjunction of literals
- B. Disjunction of variables
- C. Conjunction of literals
- D. Conjunction of variable

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Disjunction of literals

4) Rule of inference, called _____ when added to resolution principle, guarantees refutation completeness, even when involving equality. *1 point*

- A. Demodulation
- B. Paramodulation
- C. Modulation
- D. Equimodulation

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Paramodulation

5) A resolution trace captures the information of a _____ , in a linear form. *1 point*

- A. Domain
- B. Conceptualization
- C. Resolution Graph
- D. Abstraction

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. Resolution Graph

6) Answer extraction involves converting a refutation tree to a proof tree with statement at the root that can be used as an answer. Which of the following are true w.r.t the above process of answer extraction: *1 point*

I. Convert every clause arising from the negation of the goal well-formed formula into a tautology.
II. The statement at the root of the Modified Proof Tree logically follows from the axioms and the tautologies.

- A. Either I or II
- B. Both I and II
- C. I only
- D. II only

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Both I and II

7) A control strategy for a refutation system is said to be _____ if its use results in a procedure that will find a contradiction whenever one exists. *1 point*

- A. Optimal
- B. Complete
- C. Efficient
- D. Sound

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Complete

8) Under which of the following conditions, a clause {Li} subsumes a clause {Mi}, if there exists a substitution 's'? *1 point*

Condition 1: {Li}s is a subset of {Mi}
Condition 2: {Mi}s is a subset of {Li}

- A. No clause ever subsumes another clause.
- B. Condition 1 only
- C. Both Condition 1 and Condition 2
- D. Condition 2 only

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Condition 1 only

9) In this form of refutation, each resolvent has a parent that is either in the base set or that is an ancestor of the other parent. *1 point*

- A. Breadth-first Strategy
- B. Ancestry-filtered Form Strategy
- C. Linear-input Form Strategy
- D. Set-of-support Strategy

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Ancestry-filtered Form Strategy

10) In answer extraction, we convert the question to a goal well-formed formula containing an existential quantifier. The _____ represents an answer to the question. *1 point*

- A. local variable
- B. existentially quantified variable
- C. global variable
- D. well-formed formula

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. existentially quantified variable

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Fundamentals of Artificial Intelligence
MOOCs; July - Dec 2026
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Assignment No. 7: Detailed Solution

10 Marks

Each question carries 01 Mark each.

Q1. A logical agent differs from a probabilistic agent because a probabilistic agent _____.

A. uses binary truth values.
B. uses graded belief values.
C. uses deterministic rules only.
D. always produces exact answers.

Correct Answer: (B)

Q2. Within Bayesian Decision Making, probability provides a formal way of representing _____ in the presence of incomplete knowledge.

A. information
B. randomness
C. belief
D. uncertainty.

Correct Answer: (D)

Q3. Bayesian Networks assume that all direct probabilistic dependencies are explicitly represented by the network arcs. The assumption that each node is conditionally independent of its non-descendants given its parents is known as the _____.

A. Bayesian Dependency
B. Local Markov Property
C. Bayesian Property
D. Arc Consistency

Correct Answer: (B)

Q4. In a Bayesian Network, if nodes X and Y are d-separated by Z, then X and Y are _____ given Z.

A. conditionally dependent
B. conditionally independent
C. marginally independent
D. dependent

Correct Answer: (B)

Q5. A fair die is rolled once. Let event A = obtaining an even number.
What is P(A)?

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A. 1/3
B. 1/2
C. 2/3
D. 1/6
Correct Answer: (B)
Detailed Solution:
For one dice, n(S) = 6.
Events favourable for even numbers n(A) = 3

$$P(A) = \frac{n(A)}{n(S)} = \frac{3}{6} = \mathbf{\frac{1}{2}}$$

Q6. A disease affects 5% of a population.
A diagnostic test correctly identifies an infected person with probability 0.90.
What is the probability that a randomly selected individual is both infected and correctly identified?

A. 0.045
B. 0.050
C. 0.090
D. 0.950
Correct Answer: (A)
Detailed Solution:
Probability of an individual affected by the disease, P(A) = 0.05
Probability of correct identification of the disease using diagnostic test, P(B) = 0.90
So, Probability that a randomly selected individual is both infected and correctly identified,

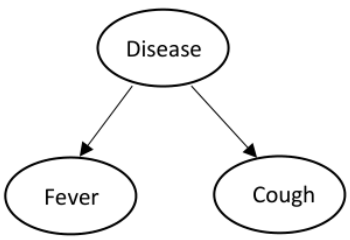
$$P(A \cap B) = P(A).P(B) = (0.05) * (0.90) = \mathbf{0.045}$$

Q7. In a class, 60% of students pass Mathematics.
Among those who pass Mathematics, 80% pass AI.
What is the probability that a randomly selected student passes both Mathematics and AI?

A. 0.48
B. 0.60
C. 0.80
D. 0.32
Correct Answer: (A)
Detailed Solution:
Probability that a randomly selected student passes both Mathematics and AI is given by

$$P(A).P(B) = 0.6 * 0.8 = \mathbf{0.48}$$

Q8. Consider the Bayesian Network shown.
Given: P(Disease = T) = 0.20
P(Fever = T| Disease = T) = 0.80
P(Fever = T| Disease = F) = 0.10



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What is the probability that a patient has fever?

A. 0.18
B. 0.30
C. 0.24
D. 0.50
Correct Answer: (C)
Detailed Solution:

$$P(Fever = T) = P(Disease = T) * P(Fever = T|Disease = T) + P(Disease = F) * P(Fever = T|Disease = F)$$

$$P(Fever = T) = 0.20 * 0.80 + (1 - 0.20) * 0.10 = \mathbf{0.24}$$

Q9. Machine A produces 60% of items with defect rate 2%.
Machine B produces 40% of items with defect rate 5%.
An item selected at random is found defective.
What is the probability that it was produced by Machine B?

A. 0.200
B. 0.625
C. 0.375
D. 0.500
Correct Answer: (B)
Detailed Solution:
Let's consider the following,
A: Item is produced by machine A
B: Item is produced by machine B
D: Item is defective
So, P(A) = 0.6; P(B) = 0.4; P(D|A) = 0.02; P(D|B) = 0.05.
Using Bayes' Theorem,

$$P(B|D) = \frac{P(D|B) * P(B)}{P(D|A) * P(A) + P(D|B) * P(B)}$$

$$P(B|D) = \frac{(0.05 * 0.40)}{(0.02 * 0.60) + (0.05 * 0.40)} = \mathbf{0.625}$$

Q10. A decision-maker has two possible actions.

Action	Utility if Success	Utility if Failure
A1	100	-20
A2	80	-5

Probability of success = 0.6
Probability of failure = 0.4

Which action should a rational agent choose?

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A. A1
B. A2
C. Both equal
D. Cannot be determined

Correct Answer: (A)

Detailed Solution:

The expected utility (EU) is given by,

$$EU(A_1) = P(S) * U(A_1, S) + P(F) * U(A_1, F)$$
$$EU(A_1) = (0.6 * 100) + [0.4 * (-20)] = 52$$

$$EU(A_2) = P(S) * U(A_2, S) + P(F) * U(A_2, F)$$
$$EU(A_2) = 0.6 * 80 + 0.4 * (-5) = 46$$

Since, $EU(A_1) > EU(A_2)$. So, a rational agent should choose action A_1.

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Fundamentals of Artificial Intelligence
Assignment Number 07: Detailed Solution
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Each question carries 01 Mark each.

Q1. Modelling with Bayesian Networks require an assumption: There are no direct dependencies in the system being modelled which are not already explicitly shown via arcs. This is referred to as the _____.

A. Bayesian Dependency
B. Markov Property
C. Bayesian Property
D. Arc Consistency

Q2. _____ provides ways and means of weighing up the desirability of goals and the likelihood of achieving them.

A. Decision Theory
B. Utility Theory
C. Probability Theory
D. Bayesian Networks

Q3. Belief networks is a powerful tool for representing and reasoning with uncertain knowledge. It is a graph in which the following holds:

A. A set of random variables makes up the nodes of the network.

B. Some nodes have a conditional probability table that quantifies the effects that the parent nodes have on the node.

C. It is a directed acyclic graph.

D. A set of directed links or arrows connects pairs of nodes.

For Q4. – Q6.

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Fundamentals of Artificial Intelligence
Assignment Number 07: Detailed Solution
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Markov Decision Process: A Markov Decision Process (MDP) refers to a stochastic decision-making process that uses a mathematical framework to model the decision-making of a dynamic system.

Q4. Which of the following statements correctly describe the components and properties of a Markov Decision Process

P. An MDP consists of a set of states, a set of actions, a reward function, and a transition probability function.

Q. The Markov property assumes that the future state depends only on the current state and action, not on the sequence of past states.

R. In an MDP, the policy defines the probability of transitioning from one state to another.

S. The goal in solving an MDP is typically to find a policy that maximizes the expected cumulative reward over time.

A. Only statements P and Q are correct.
B. Only statements P, Q, and S are correct.
C. All four statements are correct.
D. Only statements Q and R are correct.

Q5. In the context of a MDP, what does the value function $V^{\pi}(s)$ represent?

A. The probability of transitioning from state s to the next state under policy π.
B. The expected cumulative reward starting from state s, following policy π.
C. The total number of actions taken to reach a terminal state.
D. The immediate reward received after taking action $\pi(s)$ in state s.

Q6. Why is a discount factor $\gamma \in [0,1)$ used in infinite-horizon Markov Decision Processes?

A. To ensure that the transition probabilities remain valid.
B. To encourage the agent to delay rewards for long-term gain.
C. To give more importance to distant future rewards.
D. To ensure the total reward remains finite and the optimization is well-defined.

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Assignment Number 07: Detailed Solution
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For Q7. – Q 10.

Consider the joint probability distribution for random variables Rain and Wind.

	No Wind	Some Wind	Strong Wind	Storm
No Rain	0.10	0.20	0.05	0.01
Light Rain	0.05	0.10	0.15	0.04
Heavy Rain	0.05	0.10	0.10	0.05

Q7. Find the marginal probability P(Light Rain).

A. 0.33
B. 0.3
C. 0.34
D. 0.29

Q8. Find the marginal probability P(Strong Wind)

A. 0.3
B. 0.34
C. 0.29
D. 0.2

Q9. Find the conditional probability P(Heavy Rain | Strong Wind).

A. 0.333
B. 0.34
C. 0.294
D. 0.2

Q10. Find the conditional probability P(Some Wind | Light Rain).

A. 0.333
B. 0.34
C. 0.294
D. 0.2

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Assignment 7

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2019-09-18, 23:59 IST.

1) _____ provides ways and means of weighing up the desirability of goals and the likelihood of achieving them. *1 point*

- A. Decision Theory
- B. Utility Theory
- C. Probability Theory
- D. Bayesian Networks

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Utility Theory

2) Probability provides a way of summarizing the _____ that comes from our laziness and ignorance. *1 point*

- A. Joint probability distributions
- B. Randomness
- C. Belief
- D. Uncertainty

No, the answer is incorrect.
Score: 0

Accepted Answers:
D. Uncertainty

3) Causal chains (For example: Smoking causes cancer, which causes dyspnea) give rise to _____. *1 point*

- A. Conditionally dependence
- B. Conditionally independence
- C. Both of above
- D. None of above

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Conditionally independence

4) If every undirected path from a node in X to a node in Y is d-separated by E, then X and Y are _____ given E *1 point*

- A. Dependent
- B. Conditionally dependent
- C. Conditionally independent
- D. Independent

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. Conditionally independent

5) Bayesian networks allow compact specification of _____ *1 point*

- A. Joint probability distributions
- B. Propositional Logic statements
- C. Belief
- D. Conditional independence

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Joint probability distributions

6) Bayes rule can be used to _____ conditioned on one piece of evidence. *1 point*

- A. Solve queries
- B. Increase complexity of a query
- C. Decrease complexity of a query
- D. Answer probabilistic queries

No, the answer is incorrect.
Score: 0

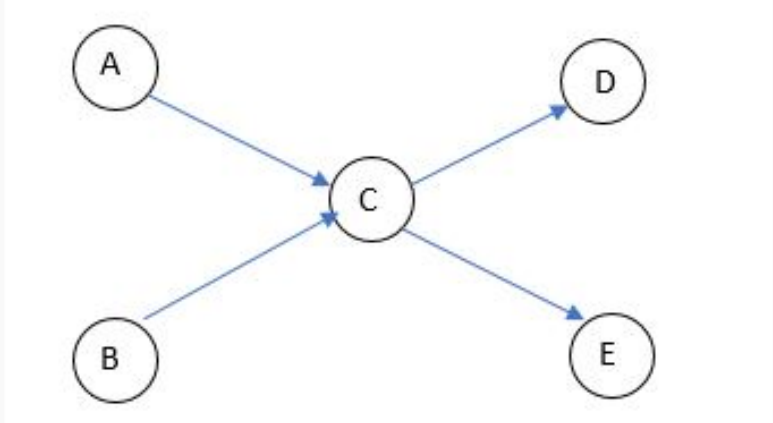
Accepted Answers:
D. Answer probabilistic queries

For Q7. – Q10. Imagine you have a party on Sunday and an exam on Monday. Too much food on Sunday could lead to lack of concentration on Monday, which would reduce the probability that you pass the exam. Another consequence of reduced concentration could be stress with your roommate as you forgot to clear the room after dinner! Lack of concentration could also be due to allergy you have after some type of food.

Consider the following random variable that can be either True or False.

A: Too much food on Sunday; B: Allergy strikes; C: Reduced concentration on Monday; D: You pass the exam; and E: Stress with your roommate.

The corresponding Bayesian Network is shown below



Given the above network, state whether the following statements are True or False

7) A and B are conditionally dependent if C or any of its descendants i.e., D or E, have received evidence. *1 point*

- True
- False

No, the answer is incorrect.
Score: 0

Accepted Answers:
True

8) E is conditionally independent of A, B and D if C is not instantiated. *1 point*

- True
- False

No, the answer is incorrect.
Score: 0

Accepted Answers:
False

9) D is conditionally independent of A, B and E if C is instantiated *1 point*

- True
- False

No, the answer is incorrect.
Score: 0

Accepted Answers:
True

10) D and E are conditionally dependent if C is not instantiated. *1 point*

- True
- False

No, the answer is incorrect.
Score: 0

Accepted Answers:
True

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Fundamentals of Artificial Intelligence
MOOCs; July - Dec 2026
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Assignment No. 8: Detailed Solution

10 Marks

Each question carries 01 Mark each.

Q1. While planning using Situation Calculus, determining which predicates (fluents) remain unchanged after an action, without explicitly specifying all unaffected predicates, gives rise to the _____.

A. Frame Problem
B. Successor-state Axiom
C. Possibility Statement
D. Frame Axioms

Correct Answer: (A)

Q2. STRIPS is an alternative representation to the pure situation calculus for planning.

I. Actions are not represented explicitly as part of the world model.
II. Actions are effectively instantaneous.
III. Actions are thought of as operators that syntactically transform world models.

Which of the above statements are correct?

A. Only Statements I and II
B. Statements I, II and III
C. Only Statements II and III
D. None.

Correct Answer: (B)

Q3. ABSTRIPS greatly reduces search space, and is more efficient at solving large problems. It is a kind of _____ planner.

A. State-space
B. Regular
C. Hierarchical
D. Advanced

Correct Answer: (C)

Q4. In a partial-order plan, a causal link from action A to action B for proposition *p* indicates that:

A. A must occur after B and deletes *p*.
B. A achieves *p* for B and protects *p*.
C. A and B are mutually exclusive.
D. *p* remains an open precondition.

Correct Answer: (B)

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Q5. The planning algorithm that breaks up a set of goal predicates into individual subgoals and attempts to solve them individually one after another in a linear order is _____

A. Forward Planning
B. Backward Planning
C. Goal Stack Planning
D. Hierarchical Task Network Planning

Correct Answer: (C)

Q6. Progression and regression are state-space search methods that build a plan as a simple list of steps, i.e., they generate _____ plans; consequently, unlike plan-space planning, they cannot take advantage of problem decomposition.

A. heuristics
B. totally ordered
C. partially ordered
D. uninformed

Correct Answer: (B)

Q7. A planning graph consists of alternating layers of:

A. States and goals.
B. Propositions and actions.
C. Actions and situations.
D. Goals and operators.

Correct Answer: (B)

Q8. Suppose an action S3 threatens a causal link S1 $\xrightarrow{p}$ S2 by deleting p. Which is a valid way to remove the threat?

A. Force S1 and S2 to occur simultaneously.
B. Delete the causal link permanently.
C. Order S3 before S1or after S2.
D. Place S3 between S1 and S2.

Correct Answer: (C)

Q9. Two actions in the same action level of a planning graph are mutex because one action deletes a precondition of the other. This is called:

A. Inconsistent effects.
B. Competing needs.
C. Interference.
D. Inconsistent support.

Correct Answer: (C)

Q10. The Sussman anomaly demonstrates a weakness of non-interleaved planning because:

A. Subgoals can always be solved independently.
B. Solving one subgoal may undo another.
C. Actions cannot have multiple preconditions.
D. Planning graphs cannot represent subgoals.

Correct Answer: (B)

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NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
Assignment Number 08: Detailed Solution
Indian Institute of Technology Guwahati

Each question carries 01 Mark each. There are MORE than ONE correct options for some of the Questions. All correct options must be identified for the answer to be evaluated as correct.

Q1. Planning algorithms use descriptions in some formal language, usually first-order logic or a subset thereof. The first key idea behind planning is to "open up " the representation of

A. States

B. Actions

C. Beliefs

D. Goals

Q2. Situation calculus is a dialect of First Order Logic. Which of the following statements hold for situation calculus?

A. Situations and actions are explicitly taken to be objects in the domain.
B. Only way to represent beliefs about a changing world.
C. Successor-state axiom specifies truth value of fluent in the next state as a function of the action and truth value in the current state.
D. All of the above.

Q3. Representing all that stays unchanged, needs to be accomplished in addition to describing what changes when an action is applied. One must find an efficient solution for stating what doesn't change. This dilemma is referred to as the _____.

A. Situation Calculus

B. Frame Problem

C. Planning Problem

D. Fluent

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Assignment Number 08: Detailed Solution
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Q4. ABSTRIPS greatly reduces search space, and is more efficient at solving large problems. It is a kind of _____ planner.

A. State-space
B. Regular
C. Hierarchical
D. Advanced

Q5. STRIPS is an alternative representation to the pure situation calculus for planning.

I. Actions are not represented explicitly as part of the world model.

II. Actions are effectively instantaneous.

III. Actions are thought of as operators that syntactically transform world models.

Which of the above statements are correct?

A. Only Statements I and II

B. Statements I, II and III

C. Only Statements II and III

D. None.

Q6. STRIPS make use of a stack that contains both subgoals and actions that have been proposed to satisfy those subgoals. It also relies on a database that describes the current situation, and a set of actions described by precondition, add and delete lists. The reasoning strategy used by STRIPS is _____

A. Database Planning
B. Goal Stack Planning
C. List Planning
D. Subgoal Planning

Q7. Progression and regression planning are _____ plan search forms. They cannot take advantage of problem decomposition.

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Fundamentals of Artificial Intelligence
Assignment Number 08: Detailed Solution
Indian Institute of Technology Guwahati

A. heuristics
B. totally ordered
C. partially ordered
D. uninformed

Q8. During the construction of the planning graph, information about the conflict between operators and literals within a layer is maintained.

I. A conflict is called a *mutex condition*, which means that a pair of literal or pair of operators is mutually exclusive.

II. For such a pair, both cannot be chosen simultaneously without leading to some kind of conflict.

Which of the above statements are correct?

A. Only Statement I

B. Both Statements I and II

C. Only Statement II

D. None.

Q9. For each layer, a *mutex relation* is defined that indicates which pairs satisfy the mutex condition. A pair, $o, o' \in O_i$, of operators is defined to be *mutex* if any of these conditions is met:

A. Inconsistent effects: An effect of o is the negated literal of an effect of o'.

B. Interference: An effect of o is the negated literal of a precondition of o'.

C. Trivial Operator: For each possible operator, o, a trivial operator is constructed for which, o' is the only precondition and effect.

D. Competing needs: A pair of preconditions, one from each of o and o', are mutex in layer L_i

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NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
Assignment Number 08: Detailed Solution
Indian Institute of Technology Guwahati

Q10. *GraphPlan* is a two-phase algorithm: Planning Graph Construction followed by Solution Extraction. Which of the following does the first stage achieve?

A. Delimiting the search space.

B. Check if a solution can be extracted.

C. Some kind of reachability analysis on the given problem.

D. The graph is treated as a constraint satisfaction problem.

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Assignment 8

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2019-09-25, 23:59 IST.

1) Planning and problem solving methods can often solve the same sorts of problems; However, planning and problem solving are different because of the ***1 point***

I. Differences in the representations of goals, states, and actions
II. Differences in the representation and construction of action sequences.
Which of the above statements are correct?

- A. None
- B. Only Statement I
- C. Both Statements I and II
- D. Only Statement II

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. Both Statements I and II

2) Type of planning which involves selecting an action whose preconditions are met until a goal state is reached. ***1 point***

- A. Regression Planning
- B. Progression Planning
- C. Plan Space Planning
- D. Graph-based Planning

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Progression Planning

3) Situation calculus is a dialect of First Order Logic. Which of the following statement holds for situation calculus? ***1 point***

- A. Situations and actions are explicitly taken to be objects in the domain.
- B. Only way to represent beliefs about a changing world.
- C. Successor-state axiom specifies truth value of fluent in the current state
- D. All of the above.

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Situations and actions are explicitly taken to be objects in the domain.

4) STRIPS is an alternative representation to the pure situation calculus for planning. ***1 point***

I. Actions are not represented explicitly as part of the world model.
II. Actions are effectively instantaneous.
III. Actions are thought of as operators that syntactically transform world models.
Which of the above statements are correct?

- A. Only Statements I and II
- B. Statements I, II and III
- C. Only Statements II and III
- D. None

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Statements I, II and III

5) A planner that can represent plans in which some steps are ordered with respect to each other and other steps are unordered is a _____ ***1 point***

- A. Linearized Plan
- B. Partial Order Plan
- C. Graph Plan
- D. Total Order Plan

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Partial Order Plan

6) _____ can solve the Sussman Anomaly. ***1 point***

- A. State-Space Planner
- B. Interleaving Planner
- C. Total-order Planner
- D. Non-interleaving Planner

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Interleaving Planner

7) Plan space planning is also referred to as _____ planning. ***1 point***

- A. Action
- B. Nonlinear
- C. Goal-stack
- D. Linear

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Nonlinear

8) In a planning graph, a pair of mutually exclusive propositions in the same layer is linked through the _____ edge. ***1 point***

- A. Mutex
- B. Negative
- C. Positive
- D. Pre-condition

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Mutex

9) GraphPlan is a two-phase algorithm; Planning Graph Construction followed by Solution Extraction. What did the first stage achieve? ***1 point***

- A. Delimiting the search space.
- B. Some kind of reachability analysis on the given problem.
- C. Both A and B above.
- D. None of A and B above

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. Both A and B above.

10) ABSTRIPS greatly reduces search space, and is more efficient at solving large problems. It is a kind of _____ planner. ***1 point***

- A. State-space
- B. Regular
- C. Hierarchical
- D. Advanced

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. Hierarchical

NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
MOOCs; July - Dec 2026
Indian Institute of Technology Guwahati
Assignment No. 9: Detailed Solution

10 Marks

Each question carries 01 Mark each.

Q1. In the context of a Markov Decision Process (MDP), which sequence best describes the basic idea of Value Iteration?

A. Compute state utilities → select optimal actions
B. Select a policy → compute its utilities → stop
C. Select actions randomly → update rewards
D. Compute rewards → construct a fixed policy
Correct Answer: (A)

Q2. In a Partially Observable Markov Decision Process (POMDP), the agent should base its decisions primarily on:

A. The most recent percept alone
B. The complete sequence of past actions
C. A probability distribution over possible states
D. A fixed sequence of predetermined actions

Correct Answer: (C)

Q3. The value of a policy is defined as the expected cumulative sum of _____ obtained while following the policy.

A. Immediate rewards
B. Discounted rewards
C. Maximum rewards
D. Terminal rewards
Correct Answer: (B)

Q4. Consider the state sequence $[S_0, S_1, S_2]$ with rewards

$$R(S_0) = 10,\ R(S_1) = 4,\ R(S_2) = 2$$

and discount factor

$$\gamma = 0.5.$$

Using the discounted-reward formulation, what is the utility of the state sequence?

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A. 12.00
B. 12.50
C. 14.00
D. 16.00

Correct Answer: (B)
Detailed Solution:
The utility of the sequence using the discounted-reward formulation,

$$U([S_0, S_1, S_2]) = R(S_0) + \gamma R(S_1) + \gamma^2 R(S_2)$$
$$U = 10 + (0.5 * 4) + (0.5^2 * 2) = \mathbf{12.50}$$

Q5. An agent considers two possible states, S_1 and S_2, with equal prior probabilities:

$$P(S_1) = P(S_2) = 0.5.$$

An agent maintains a belief distribution over possible states.
A sensor produces percept E, with:

$$P(E \mid S_1) = 0.8,\ P(E \mid S_2) = 0.2.$$

After receiving percept E, what is the updated belief

$$P(S_1 \mid E)?$$

A. 0.80
B. 0.60
C. 0.50
D. 0.20

Correct Answer: (A)
Detailed Solution:
Using Bayes' rule to find the posterior belief P(S1|E):

$$P(S_1|E) = \frac{P(E|S_1)P(S_1)}{P(E)}$$
$$P(E) = P(E|S_1)P(S_1) + P(E|S_2)P(S_2)$$
$$P(E) = (0.8 * 0.5) + (0.2 * 0.5) = 0.50$$
$$P(S_1|E) = \frac{0.40}{0.50} = \mathbf{0.80}$$

Q6. An agent in state S can choose either A_1 or A_2. The one-step action values are:

$$Q(S, A_1) = 5.2$$
$$Q(S, A_2) = 4.7.$$

Which action should the agent select under the current utility estimates?

A. Select A_1 because its action value is higher
B. Select A_2 because its action value is lower
C. Select either action because both are feasible
D. Select neither action because values are estimated

Correct Answer: (A)

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Q7. In Policy Iteration, suppose that after Policy Evaluation, the current policy $\pi_0$is found to select A_1in state S, but policy improvement shows that A_2has a higher expected utility in S.
What should the algorithm do next?

A. Retain $A_1$and terminate immediately
B. Replace $A_1$with A_2in the policy
C. Randomly select between $A_1$and A_2
D. Recalculate all transition probabilities

Correct Answer: (B)

For Q8 – Q10

Consider the following MDP: An agent is currently in state S. It can choose either A_1or A_2.
For the purposes of one-step Value Iteration, the successor states are $A$and B.

Quantity	State A	State B
Immediate reward $R(s')$	2	0
Current utility $U(s')$	4	2

The discount factor is. $\gamma = 0.5$.
The transition probabilities are:

Action	$P(A' \mid Sa)$	$P(B' \mid Sa)$
A_1	0.75	0.25
A_2	0.25	0.75

The one-step discounted action value is

$$Q(S,a) = \sum P(s' \mid S,a)[R(s') + \gamma U(s')].$$

Q8. For action A_1, what is the **expected immediate reward** over the successor states?

A. 1.50
B. 1.75
C. 2.00
D. 2.50

Correct Answer: (A)
Detailed Solution:
The expected immediate reward is

$$E[R|S,A_1] = P(A|S,A_1)R(A) + P(B|S,A_1)R(B)$$
$$E[R|S,A_1] = (0.75 * 2) + (0.25 * 0) = \mathbf{1.50}$$

Q9. Using the given reward, utility and transition values, what is the **one-step discounted action value** for A_1?

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A. 2.75
B. 3.00
C. 3.25
D. 3.50

Correct Answer: (C)
Detailed Solution:
Successor state A: R(A) = 2, U(A) = 4

$$R(A) + \gamma U(A) = 2 + 0.5(4) = 4$$

Successor state B: R(B) = 0, U(B) = 2

$$R(B) + \gamma U(B) = 0 + 0.5(2) = 1$$

Transition probabilities for A_1:

$$P(A|S, A_1) = 0.75, P(B|S, A_1) = 0.25$$

$$Q(S, A_1) = P(A|S, A_1)[R(A) + \gamma U(A)] + P(B|S, A_1)[R(B) + \gamma U(B)]$$

$$Q(S, A_1) = 0.75 * [2 + 0.5(4)] + 0.25 * [0 + 0.5(2)] = \mathbf{3.25}$$

Q10. Using the same MDP, **which action should be selected** under the current utility estimates?

A. A_1, because $Q(S, A_1) > Q(S, A_2)$
B. A_2, because $Q(S, A_2) > Q(S, A_1)$
C. Either action, because both have equal utility
D. Neither action, because the MDP is stochastic
Correct Answer: (A)

Indian Institute of Technology Guwahati

NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
Assignment Number 09: Detailed Solution
Indian Institute of Technology Guwahati

Each question carries 01 Mark each. There are MORE than ONE correct options for some of the Questions. All correct options must be identified for the answer to be evaluated as correct.

Q1. Sequential decision problems, which include utilities, uncertainty, and sensing, generalize the search and planning problems. Transition model refers to the following:

A. set of probabilities associated with the possible transitions between states after any given action.

B. a complete mapping from states to actions.

C. function be specified for the agent in order to determine the value of an action.

D. specification of the outcome probabilities for each action in each possible state.

Q2. The fundamental idea of _____ is that an agent is rational if and only if it chooses the action that yields the highest expected utility, averaged over all possible outcomes of the action.

A. Decision theory

B. Probability theory

C. Utility theory

D. Markov Decision Process

Q3. _____ provides ways and means of weighing up the desirability of goals and the likelihood of achieving them.

A. Decision Theory
B. Utility Theory
C. Probability Theory
D. Bayesian Networks

Indian Institute of Technology Guwahati

NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
Assignment Number 09: Detailed Solution
Indian Institute of Technology Guwahati

Q4. STRIPS is an alternative representation to the pure situation calculus for planning.

I. Hierarchical Plans cannot be expressed in STRIPS.

II. STRIPS operators are essentially propositional.

III. Real world projects need a better model of time than that in STRIPS.

Which of the above statements are correct?

A. Only Statements I and II

B. Statements I, II and III

C. Only Statements II and III

D. None.

Q5. _____ provide a general, concise representation for large POMDP, so they can be used as inputs for any POMDP algorithm including value and policy iteration methods.

A. Bayesian Networks

B. Decision Networks

C. Dynamic Decision Networks

D. Dynamic Belief Networks

Q6. Value Iteration is an algorithm for computing an optimal policy. The basic idea includes

A. Computing the utility of each state.

B. Use state utilities to select an optimal action in each state.

C. Calculate a new Maximum Expected Utility policy based on computed utilities.

D. Start with a random policy and calculate utilities based on if that policy were executed.

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Fundamentals of Artificial Intelligence
Assignment Number 09: Detailed Solution
Indian Institute of Technology Guwahati

Q7. Policy Iteration is an alternate algorithm for computing an optimal policy. The basic idea includes

A. Computing the utility of each state.

B. Use state utilities to select an optimal action in each state.

C. Calculate a new Maximum Expected Utility policy based on computed utilities.

D. Start with a random policy and calculate utilities based on if that policy were executed.

Q8. **Assertion A** : In real-world domains, agents have to deal with both incomplete and incorrect information.

Reason R : Incompleteness arises because the world does not necessarily match the agent's model of it.

A. Both A and R are true and R is the correct explanation for A

B. Both A and R are true but R is not the correct explanation for A

C. A is True but R is False

D. A is false but R is True

Q9. **Assertion A** : In complex real-world projects, it is common to use scheduling tools from Operations Research.

Reason R : Scheduling tools essentially take a hand constructed complete partial-order plan and generate an optimal schedule for it.

A. Both A and R are true and R is the correct explanation for A

B. Both A and R are true but R is not the correct explanation for A

C. A is True but R is False

D. A is False but R is True

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NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
Assignment Number 09: Detailed Solution
Indian Institute of Technology Guwahati

Q10. **Assertion A** : The sensor model in a belief network is the Conditional Probability Table (CPT) associated with the percept node.

Reason R : If the sensor gives a perfect report of the state, then the sensor model (the CPT) will be purely deterministic.

A. Both A and R are true and R is the correct explanation for A

B. Both A and R are true but R is not the correct explanation for A

C. A is True but R is False

D. A is false but R is True

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Assignment 9

The due date for submitting this assignment has passed. **Due on 2019-10-02, 23:59 IST.**
As per our records you have not submitted this assignment.

1) Hierarchical decomposition allows non-primitive operators to be included in plans. Which of the following statements are false? ***1 point***

- A. Hierarchical planning allows very complex plans to be built up.
- B. Primitive operators represent actions that cannot appear in the final plan.
- C. It is possible to replace a non-primitive operator with its decomposition.
- D. Primitive operators represent actions that appear in the final plan.

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Primitive operators represent actions that cannot appear in the final plan.

2) Sensor model describes how the environment generates the sensor data; in a Belief Network, it is the _____ associated with the percept node; ***1 point***
and show how reliable they are as sensors.

- A. conditional probability table
- B. maximum expected utility
- C. weighted sum of the utilities
- D. state evolution model

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. conditional probability table

3) In finding an abstract solution, and in rejecting other abstract plans as inconsistent, the downward solution property states that _____ ***1 point***

- A. If p is an abstract solution, then there is a primitive solution of which p is an abstraction.
- B. If p is an abstract solution, then there is a plan that contains abstract operators.
- C. If an abstract plan is inconsistent, then there is no primitive solution of which it is an abstraction.
- D. If p is an abstract solution, then abstract pieces can be combined hierarchically to create large plans.

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. If p is an abstract solution, then there is a primitive solution of which p is an abstraction.

4) STRIPS is an alternative representation to the pure situation calculus for planning. ***1 point***

I. Hierarchical Plans cannot be expressed in STRIPS.
II. STRIPS operators are essentially propositional.
III. Real world projects need a better model of time than that in STRIPS.

Which of the above statements are correct?

- A. Only Statements I and II
- B. Statements I, II and III
- C. Only Statements II and III
- D. None

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Statements I, II and III

5) Hierarchical Task Network (HTN) planning, uses _____ to incrementally decompose a planning problem. ***1 point***

- A. Primitive operators
- B. Abstract operators
- C. Conditional operators
- D. Hierarchical planner

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Abstract operators

6) A Markov Decision Process is a four tuple <S, A, T, R>, where S is the finite set of states, A is the finite set of actions and R is the cost or reward ***1 point***
being in state s. T is the transition model.

- A. Probability of executing action a in state s at time t
- B. Probability of s' at time t+1, given action a in state s at time t.
- C. Probability of s at time t, given actions in states upto time t-1
- D. Probability of executing action a in state s' at time t+1

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Probability of s' at time t+1, given action a in state s at time t.

7) Value Iteration is an algorithm for computing _____. ***1 point***

- A. utility of each state.
- B. an optimal policy
- C. maximum expected utility
- D. shortest path to goal

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. an optimal policy

8) For a sequential decision problem, a policy refers to _____. ***1 point***

- A. complete mapping from states to actions
- B. progressively discounted rewards of states
- C. an optimal action sequence
- D. shortest path to goal

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. complete mapping from states to actions

9) The value of a policy is the expected sum of _____ obtained, where the expectation is taken over all possible state sequences that could ***1 point***
occur, given that the policy is executed.

- A. expected utilities.
- B. additive rewards.
- C. discounted rewards.
- D. maximal rewards

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. discounted rewards.

10) _____ provide a general, concise representation for large POMDP, so they can be used as inputs for any POMDP algorithm including ***1 point***
value and policy iteration methods.

- A. Bayesian Networks
- B. Decision Networks
- C. Dynamic Decision Networks
- D. Dynamic Belief Networks

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. Dynamic Decision Networks

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Fundamentals of Artificial Intelligence
MOOCs; July - Dec 2026
Indian Institute of Technology Guwahati
Assignment No. 10: Detailed Solution

10 Marks

Each question carries 01 Mark each.

Q1. Of the two strands of Artificial Intelligence activity, the one that seeks to understand how intelligent behaviour arises is the _____.

A. cognitive approach.
B. engineering approach.
C. machine learning approach.
D. symbolic approach.

Correct Answer: (A)

Q2. A learning agent receives percepts, selects external actions, and uses feedback to improve its behaviour. Which component is primarily responsible for making improvements**?**

A. Critic
B. Learning element
C. Performance element
D. Problem generator

Correct Answer: (B)

Q3. A classifier is trained on one set of labelled examples and subsequently evaluated using previously unseen examples. The main purpose of using the unseen examples is to _____.

A. construct the training model.
B. assess the model's accuracy on new data.
C. increase the number of training attributes.
D. guarantee that the training accuracy is 100%.

Correct Answer: (B)

Q4. Which statement best distinguishes the traditional emphasis of statistics from that of machine learning?
I. Statistics often emphasizes inference, uncertainty quantification and parameter interpretation.
II. Machine learning often emphasizes predictive performance and generalization.

Choose the correct option.

A. Statements I and II
B. Only Statement II
C. Only Statement I
D. Neither I nor II

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Correct Answer: (A)

Q5. In pure inductive inference, given a collection of examples of f, return a function h that approximates f. The function h is called a _____.

A. utility function.
B. hypothesis
C. approximation function
D. classifier

Correct Answer: (B)

For Q6 – Q10

Consider the following Decision Tree learning problem. The objective is to predict the Boolean class **Play?** from the attributes **Outlook** and **Wind**.

ID	Outlook	Wind	Play?
1	Sunny	Weak	No
2	Sunny	Strong	No
3	Overcast	Weak	Yes
4	Overcast	Strong	Yes
5	Rain	Weak	Yes
6	Rain	Weak	Yes
7	Rain	Strong	No
8	Rain	Strong	No

Q6. What is the entropy of the complete training set?

A. 0.000
B. 0.500
C. 0.811
D. 1.000

Correct Answer: (D)

Detailed Solution:

$$Total\ instances\ (N) = 8; Class\ 'yes'(p) = 4; Class\ 'No'(n) = 4$$

$$Probability\ of\ each\ class: P(Yes) = \frac{4}{8} = 0.5; P(No) = \frac{4}{8} = 0.5$$

$$Entropy(H) = -0.5\log_2(0.5) - 0.5\log_2(0.5) = 0.5 + 0.5 = \mathbf{1.000}$$

Q7. If **Outlook** is used as the splitting attribute, what is the weighted entropy after the split**?**

A. 0.250
B. 0.500

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C. 0.750
D. 1.000

Correct Answer: (B)

Detailed Solution:

Sunny branch: 2 total instances and both are "No".

$$H_{Sunny} = 0.0$$

Overcast branch: 2 total instances and both are "Yes".

$$H_{Overcast} = 0.0$$

Rain branch: 4 total instances (2 are "Yes" and 2 are "No").

$$H_{Rain} = -0.5\log_2(0.5) - 0.5\log_2(0.5) = 1.0$$

$$Weighted\ entropy = \left(\frac{2}{8} * 0.0\right) + \left(\frac{2}{8} * 0.0\right) + \left(\frac{4}{8} * 1.0\right) = \mathbf{0.500}$$

Q8. What is the information gain obtained by splitting the training set on **Outlook**?

A. 0.250
B. 0.500
C. 0.750
D. 1.000

Correct Answer: (B)

Detailed Solution:

$$Information\ Gain = Total\ entropy - weighted\ entropy = 1.0 - 0.5 = \mathbf{0.500}$$

Q9. Suppose **Outlook** is selected as the root attribute. Which of the following branches produces a pure subset?

A. Only Sunny
B. Only Rain
C. Sunny and Overcast
D. None of the branches

Correct Answer: (C)

Q10. Suppose the decision tree first tests **Outlook**. For the Rain branch, it subsequently tests **Wind**. What class should the tree predict for a new instance with: **Outlook** = Rain and **Wind =** Weak?

A. Yes
B. No
C. Cannot be determined; Outlook is not Boolean
D. Cannot be determined; Training set is balanced

Correct Answer: (A)

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NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
Assignment Number 10: Detailed Solution
Indian Institute of Technology Guwahati

Each question carries 01 Mark each.

Q1. Of the two strands of Artificial Intelligence activity, the one that seeks to understand how intelligent behaviour arises is the _____.

A. cognitive approach.
B. engineering approach.
C. machine learning approach.
D. symbolic approach.

Q2. In a learning agent, the _____ informs the learning element about the performance of the action. It must use a fixed standard of performance.

A. critic
B. learning element
C. performance element
D. problem generator

Q3. Which of the following statements are correct for Machine Learning vis-à-vis Statistics?

I. Machine learning deals with Low-dimensional data [e.g. less than 100 D].

II. Huge amount of structure in the data; methods involve figuring out a way to represent this structure.

A. Statements I and II
B. Only Statement II
C. Only Statement I
D. None.

Q4. Architecture of a learning agent has FOUR components. Learning refers to the capability of autonomous acquisition and integration of knowledge. The _____ selects external actions and drives the learning element; collects percepts, decides on actions.

A. Learning element.

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Assignment Number 10: Detailed Solution
Indian Institute of Technology Guwahati

B. Performance element.
C. Problem generator.
D. Critic.

Q5. In pure inductive inference, given a collection of examples of f, return a function h that approximates f. The function h is called a _____.

A. utility function.
B. hypothesis
C. approximation function
D. classifier

For Q6 – Q10

Consider a *Decision Tree* learning problem, where properties or attributes available to describe examples in the domain are listed in the table below. Aim is to learn a definition for the goal predicate **Play?** where the definition is given as a decision tree.

ID	Outlook	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No

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Fundamentals of Artificial Intelligence
Assignment Number 10: Detailed Solution
Indian Institute of Technology Guwahati

ID	Outlook	Temperature	Humidity	Wind	Play?
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes

Q6. To implement Choose-Attribute in the DTL algorithm, we need a formal measure of "fairly good" and "really useless". _____ provides a mathematical basis for measuring the information content.

A. Information Theory
B. Utility Theory.
C. Decision Theory
D. Probability Theory

Q7. Compute the *entropy* of the class label (**Play?**) for the whole training set.

A. 0.500
B. 0.971
C. 1.000
D. 0.811

Q8. Which attribute gives the highest *information gain* when used to split the root of the tree on this dataset?

A. Temperature
B. Wind
C. Humidity
D. Outlook

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Fundamentals of Artificial Intelligence
Assignment Number 10: Detailed Solution
Indian Institute of Technology Guwahati

Q9. If the decision tree splits on **Outlook**, which branch gives a *pure subset*?

A. Sunny
B. Rainy
C. Overcast
D. None

Q10. Which of the following methods is commonly used to avoid overfitting in decision trees?

A. Increasing the depth of the tree
B. Pruning the tree
C. Removing root attributes
D. Always choosing attributes with the smallest information gain

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Assignment 10

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2019-10-09, 23:59 IST.

1) Who is recognized for developing the first machine learning program, capable of playing and winning checkers? ***1 point***

- A. John McCarthy
- B. Arthur Samuel
- C. Herbert Simon
- D. Tom M. Mitchell

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Arthur Samuel

2) The approach for modeling the relationship between a scalar dependent variable Y and one or more explanatory variables (or independent variables) denoted X is _____ ***1 point***

- A. Regression
- B. Classification
- C. Clustering
- D. Matching

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Regression

3) Of the two strands of Artificial Intelligence activity, the one that seeks to understand how intelligent behaviour arises is the _____ ***1 point***

- A. Cognitive approach
- B. Engineering approach
- C. Machine Learning approach
- D. Symbolic approach

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Cognitive approach

4) Which of the following statements are correct for Machine Learning vis-à-vis Statistics? ***1 point***

I. Machine learning deals with Low-dimensional data [e.g. less than 100 D].
II. Huge amount of structure in the data; methods involve figuring out a way to represent this structure.

- A. Statements I and II
- B. Only Statement II
- C. Only Statement I
- D. None

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Only Statement II

5) Machine learning algorithm that attempt to estimate the mapping function from the input variables to discrete or categorical output variables ***1 point***

- A. Classification
- B. Clustering
- C. Matching
- D. Regression

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Classification

6) Information content is the average information per message. Information content is also called _____. ***1 point***

- A. Common Knowledge
- B. Entropy.
- C. Knowledge
- D. Information gain

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Entropy.

7) In pure inductive inference, given a collection of examples of f, return a function h that approximates f. The function h is called a _____. ***1 point***

- A. utility function.
- B. hypothesis
- C. approximation function
- D. classifier

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. hypothesis

8) To implement Choose-Attribute in the DTL algorithm, we need a formal measure of "fairly good" and "really useless". _____ provides a mathematical basis for measuring the information content. ***1 point***

- A Information Theory
- B. Utility Theory.
- C. Decision Theory
- D. Probability Theory

No, the answer is incorrect.
Score: 0

Accepted Answers:
A Information Theory

Q9 – Q10. Architecture of a learning agent has FOUR components. Learning refers to the capability of autonomous acquisition and integration of knowledge.

9) The _____ ensures exploration; suggests actions that might lead to new experiences. ***1 point***

- A. Learning element.
- B. Performance element.
- C. Problem generator.
- D. Critic.

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. Problem generator.

10) This element selects external actions and drives the learning element; collects percepts, decides on actions. ***1 point***

- A. Learning element.
- B. Performance element.
- C. Problem generator.
- D. Critic.

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Performance element.

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MOOCs; July - Dec 2026
Indian Institute of Technology Guwahati
Assignment No. 11: Detailed Solution

10 Marks

Each question carries 01 Mark each.

Q1. Linear regression is generally not recommended for prediction outside the range of the observed data because this involves _____.

A. interpolation.
B. regularization.
C. extrapolation.
D. normalization.

Correct Answer: (C)

Q2. Consider the following statements regarding a Maximum Margin Classifier.
I. The optimal hyperplane is the one that maximizes the margin between the two classes.
II. The linear classifier defined by the maximum-margin hyperplane is called a maximum-margin classifier.
Choose the correct option.

A. Both Statements I and II
B. Only Statement II
C. Only Statement I
D. Neither Statement I nor II

Correct Answer: (A)

Q3. Support Vector Machines (SVMs) formalize binary classification as the task of finding the maximum-margin hyperplane — the linear separator that maximizes the distance to the nearest data point on each side.
For the hard-margin formulation, which one of the following best describes a support vector?

A. Any training data point.
B. Point farthest from the hyperplane.
C. Point closest to the hyperplane.
D. Mean vector of each class.

Correct Answer: (C)

Q4. SVMs can solve non-linear classification problems because both learning and classification rely only on dot products, which can be replaced by kernel functions.
Consider the following statements.
I. Learning depends only on the dot products between pairs of training samples.

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II. Classification depends only on the dot products between the unknown sample and the training samples.

A. Only Statement I
B. Only Statement II
C. Both Statements I and II
D. Neither Statement I nor II

Correct Answer: (C)

Q5. The SVM dual formulation enables nonlinear classification through the kernel trick because:
I. The dual formulation depends on training examples through inner products.
II. A kernel function can replace these inner products to implicitly operate in a higher-dimensional feature space.
A. Statement I only.
B. Statement II only.
C. Both Statement I and II.
D. Neither Statement I nor II.

Correct Answer: (C)

Q6. Clustering is a technique for finding similarity groups in data, called clusters. Identify the correct statement(s) vis-à-vis clustering.
I. Clustering is an unsupervised learning task, where class labels are not specified and only the feature vectors representing objects/instances/records are known.
II. A cluster is a collection of data items which are "similar" between them, and "dissimilar" to data items in other clusters.

A. Only Statement I.
B. Only Statement II
C. Both Statement I and II
D. Neither Statement I nor II

Correct Answer: (C)

Q7. Inter-cluster separation measures how far apart different clusters are from one another and is a reflection of _____.

A. compactness
B. similarity
C. isolation
D. cohesion

Correct Answer: (C)

Q8. Clustering is difficult to evaluate, but it is widely used in practice. Which of the following is an example of an internal cluster evaluation measure?

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A. Compactness
B. Ground-truth comparison
C. Classification accuracy
D. Alternative-solution comparison

Correct Answer: (A)

For Q9 – Q10.

Linear regression fits a line $Y = f(X)$ to a set of data points by minimizing the sum of squared vertical distances between the observed values and the fitted line. The strength of the resulting linear relationship is quantified using the correlation coefficient r and the coefficient of determination r^2.

Q9. Suppose a regression model yield $r = 0.7$.
Which of the following correctly interprets this value?

A. There is a weak negative linear relationship between X and Y.
B. There is a strong positive linear relationship between X and Y.
C. There is a strong negative linear relationship between X and Y.
D. There is no linear relationship between X and Y.

Correct Answer: (B)

Q10. If $r = -0.7$, then r^2 and its correct interpretation are:
A. 0.49; weak relationship between X and Y.
B. 0.49; explains 70% variation in Y.
C. 0.49; explains 49% variation in Y.
D. 0.49; strong relationship between X and Y.

Correct Answer: (C)

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NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
Assignment Number 11: Detailed Solution
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Each question carries 01 Mark each. There are more than one correct options for some of the Questions. All correct options must be identified for the answer to be evaluated as correct.

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Q1. *Clustering* is a technique for finding similarity groups in data, called clusters. Identify the correct statements vis-à-vis clustering.

I. In clustering problems, class labels are not specified; only the feature vectors representing different objects/instances/ records or situations are known.
II. A cluster is a collection of data items which are "similar" between them, and "dissimilar" to data items in other clusters.

A. Only statement I.
B. Only statement II.
C. Both statements I and II.
D. None of the statements.

Q2. Hierarchical clustering produce a nested sequence of clusters, a tree, also called a _____.

A. Agglomeration
B. Dendrogram
C. Concept Map
D. Taxonomy

Q3. Evaluation of a cluster based on intra-cluster cohesion measures how near the data points in a cluster are to the cluster centroid and is a reflection of _____.

A. Similarity.
B. Dissimilarity.
C. Compactness.
D. Isolation.

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Q4. Evaluation of a cluster based on inter-cluster separation means that different cluster centroids should be far away from one another and is a reflection of _____.

A. Similarity.

B. Dissimilarity.

C. Compactness.

D. Isolation.

Q5. The basic idea of *Support Vector Machine* classification is about finding two parallel hyperplanes that separate the two classes of data, so that the distance between them is as large as possible Examples closest to the hyperplane are referred to as the _____.

A. margins.
B. support vectors.
C. kernel.
D. positive samples.

Q6. Which of the following statements are correct for a Maximum Margin Classifier?

I. Best hyperplane is the one that represents the largest separation, or margin, between the two classes.

II. The linear classifier defined by the maximum-margin hyperplane is known as a maximum-margin classifier.

A. Statements I and II
B. Only Statement II
C. Only Statement I
D. None.

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Assignment Number 11: Detailed Solution
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Q7. Exclusive reliance on dot products enables to solve non-linear problems in a support vector machine because of the following:

I. Learning depends only on dot products of sample pairs.

II. Classification depends only on dot products of unknown with samples.

A. Only statement I.
B. Statements I and II.
C. Only statement II.
D. None of above.

Q8. Assertion: One of the rules of thumb for preparation of data when using ordinary *Least Squares Regression* is to remove collinearity.

Reason: Linear regression will over-fit the data when we have highly correlated input variables.

Mark the correct choice as

A. Both A and R are true and R is the correct explanation for A
B. Both A and R are true but R is not the correct explanation for A
C. A is True but R is False
D. A is false but R is True

Q9. Assertion: A cluster is a collection of data items which are "similar" between them, and "dissimilar" to data items in other clusters.

Reason: In clustering problems, class labels are not specified; only the feature vectors representing different objects/instances/ records or situations are known.

Mark the correct choice as

A. Both A and R are true and R is the correct explanation for A
B. Both A and R are true but R is not the correct explanation for A
C. A is True but R is False
D. A is false but R is True

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Assignment Number 11: Detailed Solution
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Q10. Assertion: The kernel trick is an efficient way to transform data into higher dimensions.

Reason: The kernel trick allows us to project data from a training set which isn't linearly separable into a higher dimensional space where it becomes linearly separable.

Mark the correct choice as

A. Both A and R are true and R is the correct explanation for A
B. Both A and R are true but R is not the correct explanation for A
C. A is True but R is False
D. A is false but R is True

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Assignment 11

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2019-10-16, 23:59 IST.

1) Who proposed the original Support Vector Machine algorithm? ***1 point***

- A. Patrick Winston
- B. Vladimir N. Vapnik
- C. Herbert Simon
- D. Tom M. Mitchell

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Vladimir N. Vapnik

2) In SVM classification, examples closest to the hyperplane separating the +ve from the –ve samples are the _____ ***1 point***

- A. Support vectors
- B. Outliers
- C. Marginal
- D. Boundary points

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Support vectors

3) The _____ avoids the explicit mapping to get linear learning algorithms to learn a nonlinear function or decision boundary ***1 point***

- A. Kernel function
- B. Lagrangian Multipliers
- C. Maximum margin
- D. Support vector machines

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Kernel function

4) Which of the following statements are correct for a Maximum Margin Classifier? ***1 point***

I. Best hyperplane is the one that represents the largest separation, or margin, between the two classes.
II. The linear classifier defined by the maximum-margin hyperplane is known as a maximum-margin classifier.

- A. Statements I and II
- B. Only Statement II
- C. Only Statement I
- D. None

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Statements I and II

5) Certain clustering techniques are not based on the assumption that substructures exist in the clusters. On the contrary, data may be organized in a manner that clusters have subclusters within subclusters and so on. This leads to _____ clustering ***1 point***

- A. Partitional
- B. Hierarchical
- C. Spectral
- D. Organizational

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Hierarchical

6) Hierarchical clustering produce a nested sequence of clusters, a tree, also called a _____. ***1 point***

- A. Agglomeration
- B. Dendrogram
- C. Concept Map
- D. Taxonomy

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Dendrogram

7) Data points that are very far away from other data points could be errors in the data recording or some special data points with very different values. These are called _____. ***1 point***

- A. random points.
- B. outliers
- C. approximations
- D. errors.

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. outliers

8) Hierarchical algorithms find successive clusters using previously established clusters. _____ begin with each element as a separate cluster and merge them into successively larger cluster. ***1 point***

- A Agglomerative algorithms
- B. Divisive algorithms
- C. Bayesian algorithms
- D. Partitional algorithms

No, the answer is incorrect.
Score: 0

Accepted Answers:
A Agglomerative algorithms

9) Intra-cluster cohesion measures how near the data points in a cluster are to the cluster centroid and is a reflection of _____. ***1 point***

- A. Similarity.
- B. Dissimilarity.
- C. Compactness.
- D. Isolation.

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. Compactness.

10) Exclusive reliance on dot products enables approach to solve non-linear problems in a support vector machine because of the following ***1 point***

I. Learning depends only on dot products of sample pairs.
II. Classification depends only on dot products of unknown with samples.

- A. Only statement I.
- B. Statements I and II above.
- C. Only statement II.
- D. None of above.

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Statements I and II above.

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Assignment No. 12: Detailed Solution

10 Marks

Each question carries 01 Mark each.

Q1. In pure inductive inference, given a collection of examples of a target function, the objective is to return a function that approximates the target function. The function is the _____.

A. Utility function
B. Hypothesis
C. Approximation function
D. Classifier
Correct Answer: (B)

Q2. For a Markov Decision Process (MDP), Value Iteration is primarily used to compute the _____ of each state.

A. optimal utility.
B. optimal policy.
C. transition probabilities.
D. shortest path to the goal.
Correct Answer: (A)

Q3. The value of a policy is defined as the expected cumulative sum of _____ obtained while following the policy.

A. Immediate rewards
B. Discounted rewards
C. Maximum rewards
D. Terminal rewards
Correct Answer: (B)

Q4. Which of the following best describes the key advantage of Q-learning over Value Iteration in a reinforcement learning problem?

A. Q-learning directly learns the optimal policy without considering rewards.
B. Q-learning learns state–action values without estimating the transition probabilities.
C. Q-learning requires the complete transition model of the environment a priori.
D. Q-learning learns only the immediate reward associated with each action.
Correct Answer: (B)

Q5. Which statement best distinguishes feature learning in Deep Learning (DL) from traditional Machine Learning (ML)?
A. ML automatically learns hierarchical features, whereas DL uses handcrafted features.

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B. Both traditional ML and DL always require handcrafted features.
C. DL learns features automatically and represents them hierarchically at multiple levels.
D. DL eliminates the need for a learning algorithm.
Correct Answer: (C)

For Q6 – Q10

Reinforcement Learning: Consider a reinforcement learning (RL) agent interacting with its environment through the standard state–action–reward loop. The RL agent may be built using one or more of three core components: policy, value function, and model.

Q6. The "reward hypothesis" that underlies reinforcement learning states that all goals can be described by:

A. minimization of immediate reward.
B. maximization of expected cumulative reward.
C. maximization of cumulative reward.
D. maximization of the discount factor.

Correct Answer: (B)

Q7. Which of the following best defines a "policy" in reinforcement learning?

A. A prediction of future reward for a given state.
B. A map from states to actions that defines the agent's behaviour.
C. A model of how the environment transitions between states.
D. The scalar signal returned by the environment after an action.

Correct Answer: (B)

Q8. An RL agent is in state S and can choose either a_1 or a_2. Suppose that, whenever the agent is in S and chooses a_1, the distribution of the next state and reward is the same regardless of the sequence of states and actions that led to S. Which conclusion is most appropriate?

A. The environment satisfies the Markov property with respect to state S.
B. The environment must be completely deterministic.
C. The agent must use a model-based learning algorithm.
D. The reward must be independent of the action selected.

Correct Answer: (A)

Q9. An agent has estimated

$$Q(S, a_1) = 8,\ Q(S, a_2) = 5.$$

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It chooses a_2 despite $Q(S, a_1) > Q(S, a_2)$. Which of the following best explains this behaviour?

A. Q-learning always chooses the action with the lowest Q-value.
B. The agent may be using an exploration strategy such as ϵ-greedy.
C. Q-values represent immediate rewards only.
D. The environment must be deterministic.

Correct Answer: (B)

Q10. Which of the following best describes the reward function in an RL problem?

A. A mapping of every single state visited by the agent.
B. A function specifying the reward received from states or state-action transitions.
C. The probability distribution governing state transitions and rewards.
D. A fixed schedule of actions the agent follows.

Correct Answer: (B)

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NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
Assignment Number>: Detailed Solution
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Each question carries 01 Mark each.

Q1. A Reinforcement Learning (RL) agent may include one or more of these components: Policy, Value function and Model.

Which of the following statements are true in the context of a RL agent?

Statement I. A policy is the agent's behaviour and is a map from state to action.

Statement II. Value function is a prediction of the next state.

Statement III. A model is the agent's representation of the environment and predicts what the agent will do next.

A. Statements I and II

B. Statements I and III

C. Statements II and III

D. Statements I, II and III

Q2. In reinforcement learning, rather than finding a mapping from states to state values, _____ finds a mapping from state/action pairs to values.

A. Value Iteration

B. Q-learning

C. Reinforcement function

D. Value function

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Assignment Number>: Detailed Solution
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Q3. Value Iteration is an algorithm for computing an optimal policy. The basic idea includes

A. Computing the utility of each state.

B. Use state utilities to select an optimal action in each state.

C. Calculate a new Maximum Expected Utility policy based on computed utilities.

D. Start with a random policy and calculate utilities based on if that policy were executed.

Q4. Policy Iteration is an alternate algorithm for computing an optimal policy. The basic idea includes

A. Computing the utility of each state.

B. Use state utilities to select an optimal action in each state.

C. Calculate a new Maximum Expected Utility policy based on computed utilities.

D. Start with a random policy and calculate utilities based on if that policy were executed.

Q5. Learning in a multi-layer network proceeds the same way as for perceptron; Identify which of the following statements are true for the back-propagation algorithm?

A. Essentially performs gradient-descent search on the error surface.

B. Finds a mapping from state/action pairs to values.

C. One of the simplest and most general methods used for supervised training of multilayered networks.

D. Assigns blame to individual units in the respective layers.

Q6. The layers involved in any CNN model are the convolution layers and the subsampling / pooling layers which allow the network learn filters that are specific to specific parts in an image. Which of the following are true?

I. The convolution layers help the network retain the spatial arrangement of pixels that is present in any image.

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Fundamentals of Artificial Intelligence
Assignment Number>: Detailed Solution
Indian Institute of Technology Guwahati

II. The pooling layers allow the network to summarize the pixel information.

A. Only statement I.

B. Statements I and II.

C. Only statement II.

D. None of above.

Q7. _____ offer an alternative approach to maximum likelihood estimation techniques; an unsupervised deep learning approach where two neural networks compete against each other in a zero-sum game.

A. Convolutional Neural Networks

B. Recurrent Neural Networks

C. Generative Adversarial Networks

D. Autoencoders

Q8. Deep Learning (DL) which uses either deep architectures of learning or hierarchical learning approaches is a class of Machine Learning (ML). Identify the correct statements below.

A. DL approaches do not require precisely defined features.

B. Traditional ML approaches use handcrafted engineering features by applying several feature extraction algorithms, and then apply the learning algorithms.

C. In DL, the features are learned automatically and represented hierarchically in multiple levels.

D. The DL approach is not scalable.

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Fundamentals of Artificial Intelligence
Assignment Number>: Detailed Solution
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Q9. **Assertion A:** Deep Learning models often require a large amount of labeled data and high computational resources for effective training.

Reason R: Deep Learning models learn hierarchical representations through multiple layers of abstraction.

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, but R is not the correct explanation of A.

C. A is true, but R is false.

D. A is false, but R is true.

Q10. **Assertion A**: Rosenblatt and others described multilayer feed-forward networks in the late 1950s, but concentrated their research on single-layer perceptron.

Reason R: Rosenblatt and others had difficulty of finding a sensible way to update the weights between the inputs and the hidden units.

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, but R is not the correct explanation of A.

C. A is true, but R is false.

D. A is false, but R is true.

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Assignment 12

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

Due on 2019-10-23, 23:59 IST.

1) The first computational model of a neuron that sums binary inputs and outputs 1 if the sum exceeds a certain threshold value, and otherwise outputs 0 is the _____ ***1 point***

- A. Perceptron
- B. McCulloch and Pitts Model
- C. Restricted Boltzmann Machine
- D. Multi-layer Perceptron

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. McCulloch and Pitts Model

2) A computational model of a single neuron that can only represent linearly separable functions is the _____ ***1 point***

- A. Perceptron
- B. Restricted Boltzmann Machine
- C. Autoencoder
- D. Convolutional Layer

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Perceptron

3) Major breakthrough in 2006 by Hinton lead to a fast and efficient way of training Deep Belief Networks by stacking _____ one above the other. ***1 point***

- A. Restricted Boltzmann Machine
- B. Perceptron
- C. Multi-layer Perceptron
- D. Support vector machines

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Restricted Boltzmann Machine

4) A Reinforcement Learning (RL) agent may include one or more of these components: Policy, Value function and Model. Which of the following statements are true in the context of a RL agent? ***1 point***

I. A policy is the agent's behaviour and is a map from state to action.
II. Value function is a prediction of the next state.
III. A model is the agent's representation of the environment and predicts what the agent will do next.

- A. Statements I and II
- B. Statements I and III
- C. Statements II and III
- D. Statements I, II and III

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Statements I and III

5) The "goal" of a RL system is defined using the concept of a _____ ***1 point***

- A. Policy
- B. Reinforcement function
- C. Goal predicate
- D. Value function

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Reinforcement function

6) In reinforcement learning, rather than finding a mapping from states to state values, _____ finds a mapping from state/action pairs to values. ***1 point***

- A. Value Iteration
- B. Q-learning
- C. Reinforcement function
- D. Value function

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Q-learning

7) In _____ each unit is linked only to units in the next layer; there are no links between units in the same layer, no links backward to a previous layer, and no links that skip a layer. ***1 point***

- A. Convolutional Neural Network
- B. Layered feed-forward network
- C. Recurrent Neural Network
- D. Deep Belief Network

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Layered feed-forward network

8) A key difference between traditional Machine Learning (ML) and Deep Learning (DL) is in how features are extracted. Which of the following statements are true? ***1 point***

I. Traditional ML approaches use handcrafted engineering features by applying several feature extraction algorithms, and then apply the learning algorithms.
II. In the case of DL, the features are learned automatically and are represented hierarchically in multiple levels.

- A. Statements I and II
- B. Only statement I
- C. Only statement II
- D. None of above

No, the answer is incorrect.
Score: 0

Accepted Answers:
A. Statements I and II

9) In this type of network, a loop allows information to be passed from one step of the network to the next. A _____ can be thought of as multiple copies of the same network, each network passing a message to a successor. ***1 point***

- A. Multi-layer Perceptron Network.
- B. Convolutional Neural Network.
- C. Recurrent Neural Network.
- D. Deep Belief Network.

No, the answer is incorrect.
Score: 0

Accepted Answers:
C. Recurrent Neural Network.

10) The layers involved in any CNN model are the convolution layers and the subsampling / pooling layers which allow the network learn filters that are specific to specific parts in an image. Which of the following are true? ***1 point***

I. The convolution layers help the network retain the spatial arrangement of pixels that is present in any image.
II. The pooling layers allow the network to summarize the pixel information.

- A. Only statement I.
- B. Statements I and II.
- C. Only statement II.
- D. None of above.

No, the answer is incorrect.
Score: 0

Accepted Answers:
B. Statements I and II.